

Appendix D

Problems

Problem 1

Let $\{P_{t,\tau}, \tau \in [0, 1]\}$ be the logarithm of the price of a risky asset on day $t = 1, 2, \dots, n$. The range is defined by

$$r_t = \max_{\tau \in [0,1]} P_{t,\tau} - \min_{\tau \in [0,1]} P_{t,\tau}.$$

Chou (2005) proposed the Conditional Autoregressive Range (CARR) model

$$\begin{cases} r_t &= \psi_t z_t \\ \psi_t &= \omega + \alpha r_{t-1} + \beta \psi_{t-1}, \end{cases}$$

where z_t is an iid sequence of positive random variables with mean 1, $\omega > 0$, $\alpha \geq 0$ and $\beta \geq 0$.

- INTERPRETATION

1. Give an interpretation for r_t , ψ_t and the coefficients of the model.

- STRICT STATIONARITY OF THE CARR MODEL

2. Define a function $a(x)$ such that $\psi_t = \omega + a(z_{t-1})\psi_{t-1}$.
3. Give a condition on the law of $a(z_1)$ under which ψ_t can be written as a function of $\{z_u, u < t\}$ and of the parameter $\theta = (\omega, \alpha, \beta)$.
4. Deduce a necessary and sufficient condition for the existence of a strictly stationary solution to the CARR model (give the main arguments of the proof).

- MOMENTS OF THE CARR MODEL

5. Give a necessary and sufficient condition for the existence of a non anticipative strictly stationary solution satisfying $Er_t < \infty$ (give the main arguments of the proof).
 6. Assuming the existence of a non anticipative strictly stationary solution satisfying $Er_t^2 < \infty$, give Er_t^2 as function of Ez_1^2 and θ .
 7. Introducing the semi-strong white noise $u_t = r_t - E(r_t | r_u, u < t)$, show that (r_t) satisfies an ARMA representation.
- RELATION WITH THE GARCH MODEL
8. Let (s_t) be an iid sequence, independent of the sequence (r_t) , and of uniform distribution on $\{+1, -1\}$. Show that $y_t = \sqrt{r_t}s_t$ is a GARCH process.
- ESTIMATION
9. Explain how can the CARR model be estimated by using a routine for estimating a GARCH(1,1) by Quasi-Maximum Likelihood.

SOLUTION

1. The range r_t is a kind of measure of variability of the price on day t . Since (z_t) is iid with $Ez_t = 1$, if there exists a non anticipative solution to the model (*i.e.* ψ_t is function of $\{r_u, u < t\}$, independent of z_t), the variable ψ_t is the conditional mean of r_t : $\psi_t = E(r_t \mid r_u, u < t)$. The parameter ω , α and β have interpretations similar to the parameters of a GARCH(1,1): ω is just a scaling parameter, α is a reactivity parameter to a chock of range on the previous day, β
2. Setting $a(x) = \alpha x + \beta$ we have $\psi_t = \omega + a(z_{t-1})\psi_{t-1}$.
3. Iterating the previous relation, we obtain

$$\psi_t = \omega + \omega \sum_{i=1}^{\infty} \prod_{j=1}^i a(z_{t-j})$$

provided the series with positive terms converges in $[0, +\infty[$. By the Cauchy rule, this is the case when $\gamma = E \log a(z_1) < 0$.

4. Since

$$\psi_t \geq \omega + \omega \sum_{i=1}^N \prod_{j=1}^i a(z_{t-j}) \quad \forall N \geq 1,$$

the Cauchy rule also shows that $\gamma \leq 0$ is also necessary for the existence of some finite ψ_t satisfying the equation. In the case $\gamma = 0$, the sequence $\left\{ \log \prod_{j=1}^i a(z_{t-j}) \right\}_i$ is a random walk. If this random walk is degenerated (*i.e.* $a(z_t) = 1$ a.s.) then $\psi_t = +\infty$. If this random walk is not degenerated, by the Chung-Fuchs theorem, it exceeds almost surely any arbitrarily large value, and thus we also have $\psi_t = +\infty$ a.s. The necessary and sufficient condition for strict stationarity is thus $\gamma < 0$.

5. If such a solution exists, then

$$Er_t = E\psi_t = \omega + (\alpha + \beta)Er_t.$$

This is possible only if $\alpha + \beta \geq 1$. Conversely, if $\alpha + \beta < 1$ then $\gamma < 0$ (by Jensen's inequality) and

$$E\psi_t = \omega + \omega \sum_{i=1}^{\infty} E \prod_{j=1}^i a(z_{t-j}) = \frac{\omega}{1 - \alpha - \beta}.$$

Thus $Er_t = E\psi_t$ exists if and only if $\alpha + \beta < 1$.

6. If such a solution exists, then

$$\begin{aligned} Er_t^2 &= E\psi_t^2 Ez_1^2, \\ E\psi_t^2 &= \omega^2 + \alpha^2 Er_t^2 + \beta^2 E\psi_t^2 + 2\omega(\alpha + \beta)Er_t + 2\alpha\beta E\psi_t^2. \end{aligned}$$

This entails $\alpha^2 Ez_1^2 + \beta^2 + 2\alpha\beta < 1$, and we have

$$\begin{aligned} Er_t^2 &= \frac{\omega^2 + 2\omega(\alpha + \beta)Er_t}{1 - \alpha^2 Ez_1^2 - \beta^2 - 2\alpha\beta} Ez_1^2 \\ &= \frac{\omega^2(1 + \alpha + \beta)}{(1 - \alpha^2 Ez_1^2 - \beta^2 - 2\alpha\beta)(1 - \alpha - \beta)} Ez_1^2. \end{aligned}$$

7. Since $E(r_t \mid r_u, u < t) = \psi_t$, we have the ARMA(1,1) equation

$$r_t - (\alpha + \beta)r_{t-1} = \omega + u_t - \beta u_{t-1}.$$

8. Setting $\sigma_t = \sqrt{\psi_t}$ et $\eta_t = \sqrt{r_t}s_t$, we have

$$\begin{cases} y_t = \sigma_t \eta_t \\ \sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}$$

where (η_t) is iid $(0,1)$.

9. In view of the previous question, it suffices to replace the squares of the GARCH observations by the r_t 's.

Problem 2

The exercises are independent. Let (η_t) be a sequence of iid random variables satisfying $E(\eta_t) = 0$ and $\text{Var}(\eta_t) = 1$.

Exercise 1: Consider, for all $t \in \mathbb{Z}$, the model

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t = \omega + \sum_{i=1}^q \alpha_i |\epsilon_{t-i}| + \sum_{j=1}^p \beta_j \sigma_{t-j}, \end{cases}$$

where the constants satisfy $\omega > 0$, $\alpha_i \geq 0$, $i = 1, \dots, q$ and $\beta_j \geq 0$, $j = 1, \dots, p$. We also assume that η_t is independent of the past values of ϵ_t . Let $\mu = E|\eta_t|$.

1. Give a necessary condition for the existence of $E|\epsilon_t|$, and give the value of $m = E|\epsilon_t|$.
2. In this question, assume that $p = q = 1$.
 - (a) Establish a sufficient condition for strict stationarity using the representation

$$\sigma_t = \omega + a(\eta_{t-1})\sigma_{t-1},$$

and give a strictly stationary solution of the model. It will be assumed that this condition is also necessary.

- (b) Establish a necessary and sufficient condition for the existence of a second-order stationary solution. Compute the variance of this solution.
3. Give a representation of the model which allows the coefficients to be estimated by least squares.

Exercise 2: The following parametric specifications have been introduced in the ARCH literature.

1. Quadratic GARCH(p, q):

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \left(\omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i} \right)^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2.$$

2. Qualitative ARCH of order 1:

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t = \sum_{i=1}^I \alpha_i \mathbb{1}_{A_i}(\epsilon_{t-1}),$$

where the α_i are real coefficients, $(A_i, i = 1, \dots, I)$ is a partition of $\mathbb{R}$ and $\mathbb{1}_{A_i}(x)$ is equal to 1 if $x \in A_i$, and to 0 otherwise.

3. Autoregressive stochastic volatility:

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t = \omega + \beta \sigma_{t-1} + \sigma u_t,$$

where the u_t are iid variables with mean 0 and variance 1, and are independent of $(\eta_t, t \in \mathbb{Z})$.

Briefly discuss the dynamics of the solutions (in terms of trajectories and asymmetries), the constraints on the coefficients and why each of these models is of practical interest.

In the case of model (ii), determine maximum likelihood estimators of the coefficients α_i , assuming that the A_i are known and that η_t is normally distributed.

Exercise 3: Consider the model

$$\begin{cases} \epsilon_t &= \sigma_{1t} \eta_{1t} + \sigma_{2t} \eta_{2t} \\ \sigma_{lt}^2 &= \omega_l + \sum_{i=1}^q \alpha_{li} \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_{lj} \sigma_{l,t-j}^2, \quad l = 1, 2, \end{cases}$$

where (η_{1t}, η_{2t}) is an iid sequence with values in $\mathbb{R}^2$ such that $E(\eta_{1t}) = E(\eta_{2t}) = 0$, $\text{Var}(\eta_{1t}) = \text{Var}(\eta_{2t}) = 1$ and $\text{Cov}(\eta_{1t}, \eta_{2t}) = 0$; σ_{1t} and σ_{2t} belong to the σ -field generated by the past of ϵ_t ; and $\omega_l > 0$, $\alpha_{li} \geq 0$ ($i = 1, \dots, q$), and $\beta_{lj} \geq 0$ ($j = 1, \dots, p$).

1. We assume in this question that there exists a second-order stationary solution (ϵ_t) . Show that $E\epsilon_t = 0$. Under a condition to be specified, compute the variance of ϵ_t .
2. Compute $E(\epsilon_t^2 | \epsilon_{t-1}, \epsilon_{t-2}, \dots)$ and $E(\epsilon_t^4 | \epsilon_{t-1}, \epsilon_{t-2}, \dots)$. Do GARCH-type solutions of the form $\epsilon_t = \sigma_t \eta_t$ exist, where σ_t belongs to the past of ϵ_t and η_t is independent of that past?
3. In the case where $\beta_{1j} = \beta_{2j}$ for $j = 1, \dots, p$, show that (ϵ_t) is a weak GARCH process.

SOLUTION

Exercise 1:

1. We have

$$E|\epsilon_t| = E(\sigma_t)|\eta_t| = \mu \left\{ \omega + \sum_{i=1}^q \alpha_i E|\epsilon_{t-i}| + \sum_{j=1}^p \beta_j E(\sigma_{t-j}) \right\},$$

and, putting $r = \max(p, q)$, m satisfies

$$m \left\{ 1 - \sum_{i=1}^r (\alpha_i \mu + \beta_i) \right\} = \mu \omega.$$

The condition is thus $\sum_{i=1}^r (\alpha_i \mu + \beta_i) < 1$.

2. (a) We obtain $a(\eta_{t-1}) = \alpha_1 |\eta_{t-1}| + \beta_1$. Arguing as in the GARCH(1, 1) case, it can be shown that a sufficient strict stationarity condition is $E\{\log a(\eta_t)\} < 0$.
- (b) First assume that there exists a second-order stationary solution (ϵ_t) . We have $E(\epsilon_t^2) = E(\sigma_t^2)$. Thus

$$E(\epsilon_t^2) = \omega^2 + (\alpha_1^2 + \beta_1^2 + 2\alpha_1\beta_1\mu) E(\epsilon_{t-1}^2) + 2\omega \left(\alpha_1 + \frac{\beta_1}{\mu} \right) E|\epsilon_{t-1}|$$

and, since $E(\epsilon_t^2) = E(\epsilon_{t-1}^2)$,

$$(1 - \alpha_1^2 - \beta_1^2 - 2\alpha_1\beta_1\mu) E(\epsilon_t^2) = \omega^2 + 2\omega \left(\alpha_1 + \frac{\beta_1}{\mu} \right) E|\epsilon_{t-1}|.$$

Thus we necessarily have

$$\alpha_1^2 + \beta_1^2 + 2\alpha_1\beta_1\mu < 1.$$

One can then compute

$$E|\epsilon_t| = \frac{\mu\omega}{1 - \alpha_1\mu - \beta_1}, \quad E(\epsilon_t^2) = \frac{(1 + \alpha_1\mu + \beta_1)\omega^2}{(1 - \alpha_1\mu - \beta_1)(1 - \alpha_1^2 - \beta_1^2 - 2\alpha_1\beta_1\mu)}.$$

Conversely, if this condition holds true, that is, if $E\{a^2(\eta_t)\} < 1$, by Jensen's inequality we have

$$E\{\log a(\eta_t)\} = \frac{1}{2}E\{\log a^2(\eta_t)\} \leq \frac{1}{2}\log E\{a^2(\eta_t)\} < 0.$$

Thus there exists a strictly stationary solution of the form

$$\epsilon_t = \eta_t \omega \sum_{k=0}^{\infty} a(\eta_{t-1}) \dots a(\eta_{t-k}),$$

with the convention that $a(\eta_{t-1}) \dots a(\eta_{t-k}) = 1$ if $k = 0$. Using Minkowski's inequality, it follows that

$$\begin{aligned} \sqrt{E\epsilon_t^2} &= \omega \left[E \left\{ \sum_{k=0}^{\infty} a(\eta_{t-1}) \dots a(\eta_{t-k}) \right\}^2 \right]^{1/2} \\ &\leq \omega \sum_{k=0}^{\infty} [E\{a(\eta_t)^2\}]^{k/2} < \infty. \end{aligned}$$

3. Assume that the condition of part 1 is satisfied. We have

$$E(|\epsilon_t| \mid \epsilon_u, u < t) = \mu \sigma_t.$$

Let

$$u_t = |\epsilon_t| - E(|\epsilon_t| \mid \epsilon_u, u < t)$$

be the innovation of $|\epsilon_t|$. We know that (u_t) is a noise (if $E\epsilon_t^2 < \infty$). Multiplying the equation of σ_t by μ and replacing $\mu\sigma_{t-j}$ by $|\epsilon_{t-j}| - u_{t-j}$, we obtain

$$|\epsilon_t| - \sum_{i=1}^r (\alpha_i \mu + \beta_i) |\epsilon_{t-i}| = \omega \mu + u_t - \sum_{j=1}^p \beta_j u_{t-j}.$$

This is an ARMA(r, p) equation, which can be used to estimate the coefficients by least squares.

Exercise 2:

Model (i) displays some asymmetry: positive and negative past values of the noise do not have the same impact on the volatility. Its other properties are close to the standard GARCH model: in particular, it should allow for volatility clustering. Positivity constraints are not necessary on ω and the α_i but are required for the β_j . The volatility is no longer a linear function of the past squared values, which makes its study more delicate.

Model (ii) has a constant volatility on intervals. The impact of the past values depends only on the interval of the partition to which they belong. If the α_i are well chosen, the largest values will have a stronger impact than the smallest values, and the impact could be asymmetric. If I is large, then the model is more flexible, but numerous coefficients have to be estimated (and thus it is necessary to have enough observations within each interval).

Model (iii) is a stochastic volatility model. The process σ_t cannot be interpreted as a volatility because it is not positive in general. Moreover, $|\sigma_t|$ is not exactly the volatility of ϵ_t in the sense that σ_t^2 is not the conditional variance of ϵ_t . At least when the distributions of the two noises are symmetric, the model should be symmetric and the trajectories should resemble those of a GARCH.

For model (ii), neglecting the initial value, up to a constant the log-likelihood has the form of that of the standard GARCH models:

$$L_n(\theta) = -\frac{1}{2} \sum_{t=1}^n \left\{ \log \sigma_t^2(\theta) + \frac{\epsilon_t^2}{\sigma_t^2(\theta)} \right\}.$$

Thus the first-order conditions are

$$0 = \frac{\partial L_n(\theta)}{\partial \theta} = -\frac{1}{2} \sum_{t=1}^n \frac{\partial \sigma_t^2(\theta)}{\partial \theta} \frac{1}{\sigma_t^2(\theta)} \left(1 - \frac{\epsilon_t^2}{\sigma_t^2(\theta)} \right).$$

We have $\sigma_t^2(\theta) = \alpha_i^2$ if $\epsilon_{t-1} \in A_i$. Let $T_i = \{t; \epsilon_{t-1} \in A_i\}$ and let $|T_i|$ be the cardinality of this set. Thus

$$0 = \frac{\partial L_n(\theta)}{\partial \alpha_i} = -\frac{1}{2} \sum_{t \in T_i} 2\alpha_i \frac{1}{\alpha_i^2} \left(1 - \frac{\epsilon_t^2}{\alpha_i^2} \right)$$

and finally, the maximum likelihood estimator of α_i is

$$\hat{\alpha}_i = \frac{1}{|T_i|} \sum_{t \in T_i} \epsilon_t^2.$$

Exercise 3:

1. We immediately have $E(\epsilon_t) = 0$ by the independence between the η_{it} and the past. Moreover,

$$E(\epsilon_t^2) = E(\sigma_{1t}^2 + \sigma_{2t}^2) = \omega_1 + \omega_2 + \sum_{i=1}^q (\alpha_{1i} + \alpha_{2i}) E(\epsilon_{t-i}^2) + \sum_{j=1}^p \beta_{1j} E(\sigma_{1,t-j}^2) + \beta_{2j} E(\sigma_{2,t-j}^2).$$

Thus if $\beta_{1j} = \beta_{2j} := \beta_j$ for all j , we obtain

$$E(\epsilon_t^2) = \frac{\omega_1 + \omega_2}{1 - \sum_{i=1}^q (\alpha_{1i} + \alpha_{2i}) - \sum_{j=1}^p \beta_j}.$$

2. Let $\mu_i = E(\eta_{it}^4)$. We have

$$\begin{aligned} E(\epsilon_t^2 \mid \epsilon_u, u < t) &= \sigma_{1t}^2 + \sigma_{2t}^2, \\ E(\epsilon_t^4 \mid \epsilon_u, u < t) &= \mu_1 \sigma_{1t}^4 + \mu_2 \sigma_{2t}^4. \end{aligned}$$

In general, there exists no constant k such that

$$\{E(\epsilon_t^2 \mid \epsilon_u, u < t)\}^2 = k E(\epsilon_t^4 \mid \epsilon_u, u < t),$$

which shows that (ϵ_t) is not a strong GARCH process.

3. Let $u_t = \epsilon_t^2 - \sigma_{1t}^2 - \sigma_{2t}^2$ be the innovation of ϵ_t^2 . Replacing $\sigma_{1,t-j}^2 + \sigma_{2,t-j}^2$ by $\epsilon_{t-j}^2 - u_{t-j}$, we have

$$\epsilon_t^2 = \omega_1 + \omega_2 + \sum_{i=1}^q (\alpha_{1i} + \alpha_{2i}) \epsilon_{t-i}^2 + u_t + \sum_{j=1}^p \beta_j (\epsilon_{t-j}^2 - u_{t-j}),$$

which shows that (ϵ_t^2) satisfies an ARMA $\{\max(p, q), p\}$ equation.

Problem 3

Consider the model

$$\epsilon_t = \alpha_t \epsilon_{t-1} + \omega_t,$$

where (α_t) and (ω_t) are two sequences of iid random variables with symmetric distributions such that $E(\alpha_t) = E(\omega_t) = 0$ and

$$\text{Var}(\alpha_t) = \alpha > 0, \quad \text{Var}(\omega_t) = \omega > 0,$$

and admitting moments of order 4 at least. Assume in addition that α_t and ω_t are independent of the past of ϵ_t (that is, independent of $\{\epsilon_{t-s}, s > 0\}$).

PART I: Assume that the sequences (α_t) and (ω_t) are independent.

- I.1. Verify that, in the sense of Definition 2.1, there exists an ARCH(1) solution (ϵ_t) , with $E(\epsilon_t^2) < \infty$. Show that, in general, (ϵ_t) is not an ARCH process in the strong sense (Definition 2.2).
- I.2. For $k > 0$, write ϵ_t as a function of ϵ_{t-k} and of variables from the sequences (α_t) and (ω_t) . Show that

$$E \log |\alpha_t| < 0$$

is a sufficient condition for the existence of a strictly stationary solution. Express this condition as a function of α in the case where α_t is normally distributed. (Note that if $U \sim \mathcal{N}(0, 1)$, then $E \log |U| = -0.63$.)

- I.3. If there exists a second-order stationary solution, determine its mean, $E(\epsilon_t)$, and its autocovariance function $\text{Cov}(\epsilon_t, \epsilon_{t-h})$, $\forall h \geq 0$.
- I.4. Establish a necessary and sufficient condition for the existence of a second-order stationarity solution.
- I.5. Compute the conditional kurtosis of ϵ_t . Is it different from that of a standard strong ARCH?
- I.6. Is the model stable by time aggregation?

PART II: Now assume that $\alpha_t = \lambda \omega_t$, where λ is a constant.

- II.1. Do ARCH solutions exist?

- II.2. What is the standard property of the financial series for which this model seems more appropriate than that of part I? For that property, what should be the sign of λ ?
- II.3. Assume there exists a strictly stationary solution (ϵ_t) such that $E(\epsilon_t^4) < \infty$.
- (a) Justify the equality $\text{Cov}(\epsilon_t, \epsilon_{t-h}^2) = 0$, for all $h > 0$.
 - (b) Compute the autocovariance function of (ϵ_t^2) .
 - (c) Prove that (ϵ_t) admits a weak GARCH representation.

PART III: Assume that the variables α_t and ω_t are normally distributed and that they are correlated, with $\text{Corr}(\alpha_t, \omega_t) = \rho \in [-1, 1]$. Denote the observations by $(\epsilon_1, \dots, \epsilon_n)$ and let the parameter $\theta = (\alpha, \omega, \rho)$.

- III.1. Write the log-likelihood $L_n(\theta)$ conditional on ϵ_0 .
- III.2. Solve the normal equations

$$\frac{\partial}{\partial \theta} L_n(\theta) = 0.$$

How can these equations be interpreted?

- III.3. The model is first estimated under the constraint $\rho = 0$, and then without constraint, on a series (r_t) of log-returns $(r_t = \log(p_t/p_{t-1}))$. The series contains 2000 daily observations. The estimators are assumed to be asymptotically Gaussian. The following results are obtained (the estimated standard deviations are given in parentheses):

Comment on these results. Can we accept the model of part I? Today, the price falls by 1% compared to yesterday. What are the predicted values for the returns of the next two days? How can we obtain prediction intervals?

	$\hat{\alpha}$	$\hat{\omega}$	$\hat{\rho}$	$L_n(\hat{\theta})$
Constrained model	0.54 (0.02)	0.001 (0.0002)	0	-1275.2
Unconstrained model	0.48 (0.05)	0.001 (0.0003)	-0.95 (0.04)	-1268.2

SOLUTION

PART I:

- I.1. The existence of $E(\epsilon_t^2)$ allows us to compute the first two conditional moments of ϵ_t :

$$\begin{aligned} E(\epsilon_t | \epsilon_u, u < t) &= E(\alpha_t) \epsilon_{t-1} + E(\omega_t) = 0, \\ E(\epsilon_t^2 | \epsilon_u, u < t) &= E(\omega_t^2) + E(\alpha_t^2) \epsilon_{t-1}^2 = \omega + \alpha \epsilon_{t-1}^2 := \sigma_t^2, \end{aligned}$$

in view of the independence between α_t and ω_t on one hand, and between these two variables and $\epsilon_u, u < t$, on the other hand. These conditional expectations characterize an ARCH(1) process. This process is not an ARCH in the strong sense because the variables $\epsilon_t^2 / \sigma_t^2 = \epsilon_t^2 / (\omega + \alpha \epsilon_{t-1}^2)$ are not independent (see I.5).

- I.2. We have, for $k > 0$,

$$\epsilon_t = \alpha_t \dots \alpha_{t-k+1} \epsilon_{t-k} + \omega_t + \sum_{n=1}^{k-1} \alpha_t \dots \alpha_{t-n+1} \omega_{t-n},$$

where the sum is equal to 0 if $k = 1$. Let us show that the series

$$z_t = \omega_t + \sum_{n=1}^{\infty} \alpha_t \dots \alpha_{t-n+1} \omega_{t-n}$$

is almost surely well defined. By the Cauchy root test, the series is almost surely absolutely convergent because

$$\exp \left\{ \frac{1}{n} \left(\sum_{\ell=0}^{n-1} \log |\alpha_{t-\ell}| + \log |\omega_{t-n}| \right) \right\},$$

converges a.s. to $\exp(E \log |\alpha_t|) < 1$, by the strong law of large numbers. Moreover, we have $z_t = \alpha_t z_{t-1} + \omega_t$. Thus (z_t) is a strictly stationary solution of the model. If $\alpha_t = \sqrt{\alpha} U_t$, where $U_t \sim \mathcal{N}(0, 1)$, the condition is given by $\alpha < \exp(1.26) = 3.528$.

- I.3. We have $E(\epsilon_t) = 0$, $\text{Cov}(\epsilon_t, \epsilon_{t-h}) = 0$, for all $h > 0$, and $\text{Var}(\epsilon_t) = \omega/(1 - \alpha)$.
- I.4. In view of the previous question, a necessary condition is $\alpha < 1$. To show that this condition is sufficient, note that, by Jensen's inequality, it implies the strict stationarity of the solution (z_t) . Moreover,

$$E(z_t^2) = E(\omega_t^2) + \sum_{n=1}^{\infty} E(\alpha_t^2) \dots E(\alpha_{t-n+1}^2) E(\omega_{t-n}^2) = \omega + \sum_{n=1}^{\infty} \alpha^{n-1} \omega < \infty.$$

- I.5. Assuming $E(\epsilon_t^4) < \infty$ and using the symmetry of the distributions of α_t and ω_t , we have

$$E(\epsilon_t^4 | \epsilon_u, u < t) = E(\omega_t^4) + 6\omega\alpha\epsilon_{t-1}^2 + E(\alpha_t^4)\epsilon_{t-1}^4.$$

The conditional kurtosis of ϵ_t is equal to the ratio $E(\epsilon_t^4 | \epsilon_u, u < t)/\sigma_t^2$. If this coefficient were constant we would have, for a constant K and for all t ,

$$E(\omega_t^4) + 6\omega\alpha\epsilon_{t-1}^2 + E(\alpha_t^4)\epsilon_{t-1}^4 = K(\omega + \alpha\epsilon_{t-1}^2)^2.$$

It is easy to see that this equation has no solution.

- I.6. The model satisfied by the process $(\epsilon_t^*) := (\epsilon_{2t})$ satisfies

$$\epsilon_t^* = \alpha_{2t}\alpha_{2t-1}\epsilon_{2(t-1)} + \omega_{2t} + \alpha_{2t}\omega_{2t-1} := \alpha_t^*\epsilon_{t-1}^* + \omega_t^*.$$

The independence assumptions of the initial model are not all satisfied because α_t^* and ω_t^* are not independent. The time aggregation thus holds only if a dependence between the variables ω_t and α_t is allowed in the model.

PART II:

II.1. We have

$$E(\epsilon_t^2 | \epsilon_u, u < t) = \omega(\lambda\epsilon_{t-1} + 1)^2,$$

which is incompatible with the conditional variance of an ARCH process.

II.2. The model is asymmetric because the volatility depends on the sign of ϵ_{t-1} . We should have $\lambda < 0$, so that the volatility increases more when the stock price falls than when it rises by the same magnitude.

11.3. (a) The equality follows from the independence between ω_t and all the past variables.

(b) For $h > 1$,

$$\begin{aligned} \text{Cov}(\epsilon_t^2, \epsilon_{t-h}^2) &= \text{Cov}\{\omega_t^2(\lambda\epsilon_{t-1} + 1)^2, \epsilon_{t-h}^2\} \\ &= \omega \text{Cov}(\lambda^2\epsilon_{t-1}^2 + 2\lambda\epsilon_{t-1} + 1, \epsilon_{t-h}^2) \\ &= \omega\lambda^2 \text{Cov}(\epsilon_{t-1}^2, \epsilon_{t-h}^2), \end{aligned}$$

using (a) ($\text{Cov}(\epsilon_{t-1}, \epsilon_{t-h}^2) = 0$ for $h > 1$). For $h = 1$, we obtain

$$\begin{aligned} \text{Cov}(\epsilon_t^2, \epsilon_{t-1}^2) &= \omega \text{Cov}(\lambda^2\epsilon_{t-1}^2 + 2\lambda\epsilon_{t-1} + 1, \epsilon_{t-1}^2) \\ &= \omega\lambda^2 \text{Var}(\epsilon_{t-1}^2) + 2\omega\lambda E(\epsilon_{t-1}^3). \end{aligned}$$

We have $E(\epsilon_{t-1}^3) = E(\omega_{t-1}^3) E(\lambda\epsilon_{t-2} + 1)^3 = 0$, because the distribution of ω_{t-1} is symmetric. Finally, the relation

$$\text{Cov}(\epsilon_t^2, \epsilon_{t-h}^2) = \omega\lambda^2 \text{Cov}(\epsilon_{t-1}^2, \epsilon_{t-h}^2)$$

is true for $h > 0$. For $h = 0$ we have

$$E(\epsilon_t^2) = \frac{\omega}{1 - \lambda^2\omega}, \quad E(\epsilon_t^4) = \frac{E(\omega_t^4)(6\lambda^2 E(\epsilon_t^2) + 1)}{1 - \lambda^4 E(\omega_t^4)},$$

which allows us to obtain $\text{Var}(\epsilon_t^2)$ and finally the whole autocovariance function of (ϵ_t^2) .

- (c) The recursive relation between the autocovariances of (ϵ_t^2) implies that the process is an AR(1) of the form

$$\epsilon_t^2 = \omega + \omega\lambda^2\epsilon_{t-1}^2 + u_t$$

where (u_t) is a noise. The process (ϵ_t) is thus an ARCH(1) in the weak sense.

PART III:

III.1. We have

$$L_n(\theta) = \frac{-1}{2} \left(n \log 2\pi + \sum_{t=1}^n \log \sigma_t^2 + \sum_{t=1}^n \frac{\epsilon_t^2}{\sigma_t^2} \right),$$

where

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + 2\rho\sqrt{\alpha\omega}\epsilon_{t-1}.$$

III.2. The normal equations are

$$\sum_{t=1}^n \left(1 - \frac{\epsilon_t^2}{\sigma_t^2} \right) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} = 0,$$

with

$$\frac{\partial \sigma_t^2}{\partial \theta} = \left(\epsilon_{t-1}^2 + \rho\epsilon_{t-1}\sqrt{\omega/\alpha}, 1 + \rho\epsilon_{t-1}\sqrt{\alpha/\omega}, 2\sqrt{\alpha\omega}\epsilon_{t-1} \right)'.$$

These equations can be interpreted, for n large, as orthogonality relations.

- III.3. Note that the estimated coefficients are significant at the 5% level. The constrained model (and thus that of part I) is rejected by the likelihood ratio test. The sign of $\hat{\rho}$ is what was expected. The optimal prediction is 0, regardless of the horizon. We have $E(\epsilon_t^2 | \epsilon_u, u < t) = \sigma_t^2$ and $E(\epsilon_{t+1}^2 | \epsilon_u, u < t) = \alpha\sigma_t^2 + \omega$. The estimated volatility for the next day is $\hat{\sigma}_t^2 = 0.48(\log 0.99)^2 + 0.001 - 2 \times 0.95 \times \sqrt{0.48 \times 0.001} \times \log 0.99 = 0.0015$. The 95% confidence intervals are thus:

$$\text{at horizon 1, } [-1.96\hat{\sigma}_t; 1.96\hat{\sigma}_t] = [-0.075; 0.075];$$

$$\text{at horizon 2, } [-1.96\sqrt{\hat{\alpha}\hat{\sigma}_t^2 + \hat{\omega}}; 1.96\sqrt{\hat{\alpha}\hat{\sigma}_t^2 + \hat{\omega}}] = [-0.081; 0.081].$$

Problem 4

Let (η_t) be a sequence of iid random variables such that $E(\eta_t) = 0$ and $\text{Var}(\eta_t) = 1$. Consider, for all $t \in \mathbb{Z}$, the model

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t^r = \omega + \sum_{i=1}^q \alpha_i |\epsilon_{t-i}|^r + \sum_{j=1}^p \beta_j \sigma_{t-j}^r, \end{cases} \quad (\text{D.1})$$

where r is a positive real. The other constants satisfy:

$$\omega > 0, \quad \alpha_i \geq 0, \quad i = 1, \dots, q, \quad \beta_j \geq 0, \quad j = 1, \dots, p. \quad (\text{D.2})$$

1. Assume in this question that $p = q = 1$.

- Establish a sufficient condition for strict stationarity and give a strictly stationary solution of the model.
- Give this condition in the case where $\beta_1 = 0$ and compare the conditions corresponding to the different values of r .
- Establish a necessary and sufficient condition for the existence of a nonanticipative strictly stationary solution such that $E|\epsilon_t|^{2r} < \infty$. Compute this expectation.
- Assume in this question that $r < 0$. What might be the problem with that specification?

r	Estimated model
0.5	$ \epsilon_t ^{0.5} = 0.0002 + 0.4 \epsilon_{t-1} ^{0.5} + v_t - 0.5v_{t-1} - 0.2v_{t-2}$
1	$ \epsilon_t = 0.00024 + 0.5 \epsilon_{t-1} + 0.2 \epsilon_{t-2} + v_t - 0.3v_{t-1}$
1.5	$ \epsilon_t ^{1.5} = 0.00024 - 0.5 \epsilon_{t-1} ^{1.5} + v_t$
2	$\epsilon_t^2 = 0.00049 + 0.1\epsilon_{t-1}^2 + v_t - 0.3v_{t-1}$

2. Propose an extension of the model that could take into account the typical asymmetric property of the financial series.
3. Show that from a solution (ϵ_t) of (D.1), one can define a solution (ϵ_t^*) of a standard GARCH model (with $r = 2$).
4. Show that if $E|\epsilon_t|^{2r} < \infty$, then $(|\epsilon_t|^r)$ is an ARMA process whose orders will be given.
5. Assume that we observe a series (ϵ_t) of log-returns $(\epsilon_t = \log(p_t/p_{t-1}))$. For different powers of $|\epsilon_t|$, ARMA models are estimated by least squares. The orders of these ARMA models are identified by information criteria. We obtain the following results, v_t denoting a noise.

What values of r are compatible with model (D.1)–(D.2)? Assuming that the distribution of η_t is known, what are the corresponding parameters α_i and β_j ?

6. In view of the previous questions discuss the interest of the models defined by (D.1).

SOLUTION

1. (a) We have

$$\sigma_t^r = \omega + \alpha_1 |\epsilon_{t-1}|^r + \beta_1 \sigma_{t-1}^r = \omega + (\alpha_1 |\eta_{t-1}|^r + \beta_1) \sigma_{t-1}^r := \omega + a(\eta_{t-1}) \sigma_{t-1}^r.$$

Thus, for $N > 0$,

$$\sigma_t^r = \omega \left\{ 1 + \sum_{n=1}^N a(\eta_{t-1}) \dots a(\eta_{t-n}) \right\} + a(\eta_{t-1}) \dots a(\eta_{t-N-1}) \sigma_{t-N-1}^r.$$

Proceeding as for the standard GARCH(1, 1) model, using the Cauchy root test, it is shown that if

$$E \log \{a(\eta_t)\} < 0$$

the process (h_t) , defined by

$$h_t = \lim_{N \rightarrow \infty} \text{a.s.} \left\{ 1 + \sum_{n=1}^N a(\eta_{t-1}) \dots a(\eta_{t-n}) \right\} \omega,$$

exists, takes real positive values, is strictly stationary and satisfies $h_t = \omega + a(\eta_{t-1})h_t$. A strictly stationary solution of the model is obtained by $\epsilon_t = h_t^{1/r} \eta_t$. This solution is nonanticipative, because ϵ_t is function of η_t and of its past.

- (b) If $\beta_1 = 0$, the condition is given by

$$\alpha_1 < e^{-rE \log |\eta_t|}.$$

Using the Jensen inequality, it can be seen that

$$E \log |\eta_t| = E \left\{ \frac{1}{2} \log |\eta_t|^2 \right\} \leq \frac{1}{2} \log E \{ |\eta_t|^2 \} = 0.$$

The conditions are thus less restrictive when r is larger.

- (c) If (ϵ_t) is nonanticipative and stationary and admits a moment of order $2r$, we have $E(\epsilon_t^{2r}) = E(\sigma_t^{2r})E(\eta_t^{2r})$ and, since η_{t-1} and σ_{t-1} are independent,

$$\begin{aligned} E(\sigma_t^{2r}) &= E \{ \omega + a(\eta_{t-1}) \sigma_{t-1}^r \}^2 \\ &= \omega^2 + 2\omega E \{ a(\eta_{t-1}) \} E(\sigma_{t-1}^r) + E \{ a(\eta_{t-1})^2 \} E(\sigma_{t-1}^{2r}). \end{aligned}$$

Thus, by stationarity,

$$[1 - E\{a(\eta_t)^2\}] E(\sigma_t^{2r}) = \omega^2 + 2\omega E\{a(\eta_t)\} E(\sigma_t^r) > 0.$$

A necessary condition is thus $E\{a(\eta_t)^2\} < 1$. Conversely, if this condition is satisfied we have

$$E(h_t^2) \leq 2\omega^2 \sum_{n=1}^{\infty} [E\{a(\eta_t)^2\}]^n < \infty.$$

The stationary solution that was previously given thus admits a moment of order $2r$. Using the previous calculation, we obtain

$$E(\epsilon_t^{2r}) = E(\eta_t^{2r}) \left(\frac{\omega^2}{1 - E\{a(\eta_t)^2\}} + \frac{2\omega^2 E\{a(\eta_t)\}}{[1 - E\{a(\eta_t)^2\}][1 - E\{a(\eta_t)\}]} \right).$$

- (d) If $r < 0$, $a(\eta_t)$ is not defined when $\eta_t = 0$. This is the same for the strictly stationary solution, when one of the variables η_{t-j} is null. However, the probability of such an event is null if $P(\eta_t = 0) = 0$.

2. The specification

$$\sigma_t^r = \omega + \sum_{i=1}^q (\alpha_{i,+} \mathbb{1}_{\epsilon_{t-i} > 0} + \alpha_{i,-} \mathbb{1}_{\epsilon_{t-i} < 0}) |\epsilon_{t-i}|^r + \sum_{j=1}^p \beta_j \sigma_{t-j}^r,$$

with $\alpha_{i,+} > 0$ and $\alpha_{i,-} > 0$, induces a different impact on the volatility at time t for the positive and negative past values of ϵ_t . For $\alpha_{i,+} = \alpha_{i,-}$ we retrieve model (1).

3. Let $\epsilon_t^* = |\epsilon_t|^{r/2} \nu_t$, where (ν_t) is an iid process, independent of (ϵ_t) and taking the values -1 and 1 with probability $1/2$. We have $\epsilon_t^* = \sigma_t^* \eta_t^*$ with

$$\sigma_t^{*2} = \sigma_t^r = \omega + \sum_{i=1}^q \alpha_i |\epsilon_{t-i}|^{*2} + \sum_{j=1}^p \beta_j \sigma_{t-j}^{*2},$$

and $(\eta_t^*) = (|\eta_t|^{r/2} \nu_t)$ is an iid process with mean 0 and variance 1. The process (ϵ_t^*) is thus a standard GARCH.

4. The innovation of $|\epsilon_t|^r$ is defined by

$$u_t = |\epsilon_t|^r - E(|\epsilon_t|^r | \epsilon_{t-1}, \epsilon_{t-2}, \dots) = |\epsilon_t|^r - \sigma_t^r \mu_r$$

with $E|\eta_t|^r = \mu_r$. It follows that

$$|\epsilon_t|^r = \omega\mu_r + \sum_{i=1}^{\max(p,q)} (\alpha_i\mu_r + \beta_i)|\epsilon_{t-i}|^r + u_t - \sum_{j=1}^p \beta_j u_{t-j}$$

and $(|\epsilon_t|^r)$ is an ARMA $\{\max(p, q), p\}$ process.

5. Noting that the order and coefficients of the AR part in the ARMA representation are greater (in absolute value) than those of the MA part, we see that only the model for $r = 1$ is compatible with the class of model that is considered here. We have $r = p = 1$ and $q = 2$, and the estimated coefficients are

$$\hat{\beta}_1 = 0.3, \quad \hat{\alpha}_1 = \frac{0.5 - \hat{\beta}_1}{\mu_1} = \frac{0.2}{\mu_1}, \quad \hat{\alpha}_2 = 0.2.$$

6. The proposed class can take into account the same empirical properties as the standard GARCH models, but is more flexible because of the extra parameter r .

Problem 5

Let (η_t) be a sequence of iid random variables such that $E(\eta_t) = 0$, $\text{Var}(\eta_t) = 1$, $E(\eta_t^4) = \mu_4$. Let (a_t) be another sequence of iid random variables, independent of the sequence (η_t) , taking the values 0 and 1 and such that

$$P(a_t = 1) = p, \quad P(a_t = 0) = 1 - p, \quad 0 \leq p \leq 1.$$

Consider, for all $t \in \mathbb{Z}$, the model

$$\epsilon_t = \{\sigma_{1t}a_t + \sigma_{2t}(1 - a_t)\}\eta_t, \quad (\text{D.3})$$

$$\sigma_{1t}^2 = \omega_1 + \alpha_1\epsilon_{t-1}^2, \quad \sigma_{2t}^2 = \omega_2 + \alpha_2\epsilon_{t-1}^2, \quad (\text{D.4})$$

where

$$\omega_i > 0, \quad \alpha_i \geq 0, \quad i = 1, 2.$$

A solution such that ϵ_t is independent of the future variables η_{t+h} and a_{t+h} , $h > 0$, is called nonanticipative.

1. What are the values of the parameters corresponding to the standard ARCH(1)? What kind of trajectories can be obtained with that specification?
2. In order to obtain a strict stationarity condition, write (D.4) in the form

$$Z_t := \begin{pmatrix} \sigma_{1t}^2 \\ \sigma_{2t}^2 \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + A_{t-1} \begin{pmatrix} \sigma_{1,t-1}^2 \\ \sigma_{2,t-1}^2 \end{pmatrix} \quad (\text{D.5})$$

where A_{t-1} is a matrix depending on $\eta_{t-1}, a_{t-1}, \alpha_1, \alpha_2$.

3. Deduce a strict stationarity condition for the process (Z_t) , and then for the process (ϵ_t) , as function of the Lyapunov coefficient

$$\gamma = \lim_{t \rightarrow \infty} \text{a.s.} \frac{1}{t} \log \|A_t A_{t-1} \dots A_1\|$$

(justify the existence of γ and briefly outline the steps of the proof). Note that A_t is the product of a column vector by a row vector. Deduce the following simple expression for the strict stationarity condition:

$$\alpha_1^p \alpha_2^{1-p} < c,$$

for a constant c that will be specified. How can the condition be interpreted?

4. Give a necessary condition for the existence of a second-order and nonanticipative stationary solution. It can be assumed that this condition implies strict stationarity. Deduce that the necessary second-order stationarity condition is also sufficient. Compute the variance of ϵ_t .
5. We now consider predictions of future values of ϵ_t and of its square. Give an expression for $E(\epsilon_{t+h}|\epsilon_{t-1}, \epsilon_{t-2}, \dots)$ and $E(\epsilon_{t+h}^2|\epsilon_{t-1}, \epsilon_{t-2}, \dots)$, for $h > 0$, as a function of ϵ_{t-1} .
6. What is the conditional kurtosis of ϵ_t ? Is there a standard ARCH solution to the model?
7. Assuming that the distribution of η_t is standard normal, write down the likelihood of the model. For a given series of 2000 observations, a standard ARCH(1) model is estimated, and then model (D.3)–(D.4) is estimated. The estimators are assumed to be asymptotically Gaussian. The results are presented in the following table (the estimated standard deviations are given in parentheses, $L_n(\cdot)$ denotes the likelihood):

	$\hat{\omega}_1$	$\hat{\alpha}_1$	$\hat{\omega}_2$	$\hat{\alpha}_2$	$\hat{p}$	$\log L_n(\hat{\theta})$
ARCH(1)	0.002 (0.001)	0.6 (0.2)	—	—	—	–1275.2
Model (D.3)–(D.4)	0.001 (0.001)	0.10 (0.03)	0.005 (0.000)	1.02 (0.23)	0.72 (0.12)	–1268.2

Comment on these results. Can we accept the general model? (we have $P\{\chi^2(3) > 7.81\} = 0.05$).

8. Discuss the estimation of the model by OLS.

SOLUTION

1. The standard ARCH(1) is obtained for $\alpha_1 = \alpha_2, \forall p$. It is also obtained for $p = 0, \forall \alpha_1, \alpha_2$ and for $p = 1, \forall \alpha_1, \alpha_2$. The trajectories may display abrupt changes of volatility (for instance, if ω_1 and ω_2 are very different).
2. We obtain equation (D.5) with

$$A_{t-1} = \begin{pmatrix} \alpha_1 \eta_{t-1}^2 a_{t-1}^2 & \alpha_1 \eta_{t-1}^2 (1 - a_{t-1}^2) \\ \alpha_2 \eta_{t-1}^2 a_{t-1}^2 & \alpha_2 \eta_{t-1}^2 (1 - a_{t-1}^2) \end{pmatrix}.$$

3. The existence of γ requires $E \log^+ \|A_t\| < \infty$. This condition is satisfied because $E\|A_t\| < \infty$, for example with the norm defined by $\|A\| = \sum |a_{ij}|$. The strict stationarity condition $\gamma < 0$ is shown as for the standard GARCH. Under this condition the strictly stationary solution of (D.5) is given by

$$Z_t = \left(I + \sum_{i=0}^{\infty} A_t A_{t-1} \dots A_{t-i} \right) \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}.$$

Note that

$$A_t = \begin{pmatrix} \alpha_1 \eta_t^2 \\ \alpha_2 \eta_t^2 \end{pmatrix} \begin{pmatrix} a_t^2 & 1 - a_t^2 \end{pmatrix}.$$

Thus

$$A_t A_{t-1} = \eta_t^2 \eta_{t-1}^2 \{ \alpha_1 a_t^2 + \alpha_2 (1 - a_t^2) \} \begin{pmatrix} \alpha_1 \eta_t^2 \\ \alpha_2 \eta_t^2 \end{pmatrix} \begin{pmatrix} a_{t-1}^2 & 1 - a_{t-1}^2 \end{pmatrix},$$

$$A_t A_{t-1} \dots A_1 = \eta_1^2 \prod_{i=0}^{t-2} \eta_{t-i}^2 \{ \alpha_1 a_{t-i}^2 + \alpha_2 (1 - a_{t-i}^2) \} \begin{pmatrix} \alpha_1 \eta_t^2 \\ \alpha_2 \eta_t^2 \end{pmatrix} \begin{pmatrix} a_0^2 & 1 - a_0^2 \end{pmatrix},$$

$$\|A_t A_{t-1} \dots A_1\| = \eta_1^2 \prod_{i=0}^{t-2} \eta_{t-i}^2 \{ \alpha_1 a_{t-i}^2 + \alpha_2 (1 - a_{t-i}^2) \} \left\| \begin{pmatrix} \alpha_1 \eta_t^2 \\ \alpha_2 \eta_t^2 \end{pmatrix} \begin{pmatrix} a_0^2 & 1 - a_0^2 \end{pmatrix} \right\|,$$

because α_1 and α_2 are positive. It follows that, by the strong law of large numbers,

$$\begin{aligned}
\frac{1}{t} \log \|A_t A_{t-1} \dots A_1\| &= \frac{1}{t} \sum_{i=0}^{t-1} \log \eta_{t-i}^2 + \frac{1}{t} \sum_{i=0}^{t-2} \log \{\alpha_1 a_{t-i}^2 + \alpha_2 (1 - a_{t-i}^2)\} \\
&\quad + \frac{1}{t} \log \left\| \begin{pmatrix} \alpha_1 \eta_t^2 \\ \alpha_2 \eta_t^2 \end{pmatrix} \begin{pmatrix} a_0^2 & 1 - a_0^2 \end{pmatrix} \right\| \\
&\rightarrow E \log \eta_t^2 + E \log \{\alpha_1 a_t^2 + \alpha_2 (1 - a_t^2)\}
\end{aligned}$$

almost surely as $t \rightarrow \infty$. The second expectation is equal to

$$p \log \alpha_1 + (1 - p) \log \alpha_2 = \log \alpha_1^p \alpha_2^{1-p}.$$

It follows that

$$\gamma < 0 \iff \alpha_1^p \alpha_2^{1-p} < \exp \{-E(\log \eta_t^2)\}.$$

We note that the condition is satisfied even if one of the coefficients, for instance α_1 , is large, provided that the corresponding probability, p , is not too large.

4. If (ϵ_t) is second-order stationary, we have

$$E\epsilon_t^2 = p \{\omega_1 + \alpha_1 E(\epsilon_t^2)\} + (1 - p) \{\omega_2 + \alpha_2 E(\epsilon_t^2)\}.$$

The necessary condition is thus

$$\bar{\alpha} := p\alpha_1 + (1 - p)\alpha_2 < 1,$$

and we have

$$\text{Var}(\epsilon_t) = \frac{p\omega_1 + (1 - p)\omega_2}{1 - p\alpha_1 - (1 - p)\alpha_2}.$$

Conversely, suppose that this condition is satisfied and that it implies that $\gamma < 0$. We have

$$E(A_t) = \begin{pmatrix} \alpha_1 p & \alpha_1 (1 - p) \\ \alpha_2 p & \alpha_2 (1 - p) \end{pmatrix}.$$

This matrix has rank 1, and thus admits a zero eigenvalue and a nonzero eigenvalue that is equal to its trace, $\bar{\alpha}$. This coefficient, less than 1 by assumption, is also the spectral radius of $E(A_t)$. It follows that the expectation of the stationary solution Z_t defined above is finite because $E A_t A_{t-1} \dots A_{t-i} = \{E(A_t)\}^{i+1}$.

5. We clearly have $E(\epsilon_{t+h} \mid \epsilon_{t-1}, \epsilon_{t-2}, \dots) = 0$. Since $a_t(1 - a_t) = 0$ we have

$$\begin{aligned}\epsilon_{t+h}^2 &= [\{\omega_1 a_{t+h}^2 + \omega_2(1 - a_{t+h}^2)\} + \{\alpha_1 a_{t+h}^2 + \alpha_2(1 - a_{t+h}^2)\} \epsilon_{t+h-1}^2] \eta_{t+h}^2 \\ &:= \tilde{\omega}_{t+h} + \tilde{\alpha}_{t+h} \epsilon_{t+h-1}^2 \\ &= \tilde{\omega}_{t+h} + \tilde{\alpha}_{t+h} \tilde{\omega}_{t+h-1} + \dots + \tilde{\alpha}_{t+h} \dots \tilde{\alpha}_{t+1} \tilde{\omega}_t + \tilde{\alpha}_{t+h} \dots \tilde{\alpha}_t \epsilon_{t-1}^2.\end{aligned}$$

Letting $\bar{\omega} = E\tilde{\omega}_t$ and since $\bar{\alpha} = E\tilde{\alpha}_t$, it follows that

$$E(\epsilon_{t+h}^2 \mid \epsilon_{t-1}, \dots) = \bar{\omega}(1 + \bar{\alpha} + \dots + \bar{\alpha}^h) + \bar{\alpha}^{h+1} \epsilon_{t-1}^2$$

for $h > 0$.

6. The conditional kurtosis is equal to

$$\frac{E(\epsilon_t^4 \mid \epsilon_{t-1}, \dots)}{\{E(\epsilon_t^2 \mid \epsilon_{t-1}, \dots)\}^2} = \frac{E(\eta_t^4)}{\{E(\eta_t^2)\}^2} \frac{p\sigma_{1t}^4 + (1-p)\sigma_{2t}^4}{\{p\sigma_{1t}^2 + (1-p)\sigma_{2t}^2\}^2}.$$

This coefficient depends on t in general, which shows that there is no standard GARCH solution to this model (except in the cases mentioned in part 1).

7. The conditional density of ϵ_t is written as

$$l_t = p \frac{1}{\sqrt{2\pi}\sigma_{1t}} \exp\left\{-\frac{\epsilon_t^2}{2\sigma_{1t}^2}\right\} + (1-p) \frac{1}{\sqrt{2\pi}\sigma_{2t}} \exp\left\{-\frac{\epsilon_t^2}{2\sigma_{2t}^2}\right\}$$

and the log-likelihood of the sample is the product of the l_t for $t = 1, \dots, n$. The estimation results display very different estimated coefficients α_1 and α_2 . Moreover, the likelihood ratio test is equivalent to comparing the difference of the log-likelihoods and the quantile of order $1 - \alpha$ of a $\chi^2(3)$. Since we have $2 \times (1275.2 - 1268.2) > 7.81$, the standard ARCH(1) is rejected in favor of the general model at the 5% level.

Problem 6

Let (η_t) be a sequence of iid random variables, such that $E(\eta_t) = 0$. When $E|\eta_t|^m < \infty$, denote $\mu_m = E\eta_t^m$. Consider the model

$$\epsilon_t = \eta_t + b\eta_t\epsilon_{t-1}, \quad t \in \mathbb{Z}. \quad (\text{D.6})$$

1. Strict stationarity

(a) Let

$$Z_{t,n} = \eta_t + \sum_{i=1}^n b^i \eta_t \eta_{t-1} \cdots \eta_{t-i}.$$

Show that if $E \log |b\eta_t| < 0$ then the sequence $(|Z_{t,n}|)_{n \geq 1}$ converges almost surely. Under this condition, let

$$Z_t = \eta_t + \sum_{i=1}^{\infty} b^i \eta_t \eta_{t-1} \cdots \eta_{t-i}.$$

(b) Show that if $E \log |b\eta_t| < 0$ then equation (D.6) admits a nonanticipative and ergodic strictly stationary solution.

(c) We have

$$\int_{-\infty}^{\infty} \log |x| \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2} \right\} dx = -0.635181.$$

Give the strict stationarity condition when $\eta_t \sim \mathcal{N}(0, \mu_2)$.

2. Second-order stationarity

(a) Under what condition is $(Z_{t,n})_n$ a Cauchy sequence in L^2 ?

(b) Show that $b^2\mu_2 < 1$ entails $E \log |b\eta_t| < 0$.

(c) Show that if $b^2\mu_2 < 1$ then $(\epsilon_t) = (Z_t)$ is the second-order stationary solution of (D.6).

(d) Assume that $\mu_2 \neq 0$. Show that the condition $b^2\mu_2 < 1$ is also necessary for the existence of a nonanticipative second-order stationary solution.

3. Properties of the marginal distribution and conditional moments

Assume that $b^2\mu_2 < 1$ and that (ϵ_t) is the second-order stationary solution of (D.6).

- (a) Show that (ϵ_t) is a weak GARCH process whose orders will be specified.
- (b) Compare the conditional variance of (ϵ_t) with that of a strong ARCH(1). Does the sign of ϵ_{t-1} have an impact on the volatility at time t ? Is this property of interest for financial series?
4. **Estimation** Denote by b_0 and μ_{02} the true value of the parameters b and μ_2 . Assume that $b_0^2\mu_{02} < 1$ and that $\epsilon_1, \dots, \epsilon_n$ is a second-order stationary realization of model (D.6). Let

$$h_t = h_t(b, \mu_2) = \mu_2(1 + b\epsilon_{t-1})^2 \quad \text{and} \quad h_{0t} = h_t(b_0, \mu_{02}).$$

- (a) What is the interpretation of $\nu_t = \epsilon_t^2 - h_{0t} = (\eta_t^2 - \mu_{02})(1 + b_0\epsilon_{t-1})^2$?
- (b) Assume that $E\epsilon_t^4 < \infty$. Show that, almost surely,

$$\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{t=2}^n \nu_t (h_{0t} - h_t) = 0.$$

- (c) Assume that the distribution of ϵ_t is not concentrated on one or two points (in particular, $\mu_2 \neq 0$). Show that

$$E(h_{0t} - h_t)^2 = 0 \quad \text{if and only if} \quad b = b_0 \text{ and } \mu_2 = \mu_{02}.$$

- (d) Under the previous assumptions, consider the criterion

$$Q_n(b, \mu_2) = \frac{1}{n} \sum_{t=2}^n \{\epsilon_t^2 - h_t\}^2.$$

Show that, almost surely,

$$\lim_{n \rightarrow \infty} Q_n(b, \mu_2) \geq \lim_{n \rightarrow \infty} Q_n(b_0, \mu_{02})$$

with equality if and only if $b = b_0$ and $\mu_2 = \mu_{02}$. Give an estimation method for the parameters.

- (e) Describe the quasi-maximum likelihood method.

5. **Extension** Without giving detailed proofs, extend the stationarity and estimation results to the model

$$\epsilon_t = \eta_t + b_1\eta_t\epsilon_{t-1} + \dots + b_q\eta_t\epsilon_{t-q}, \quad t \in \mathbb{Z}.$$

6. **Further extensions** Assume that $\mu_2 \neq 0$, $\mu_3 = 0$ and $\mu_4 b^4 < 1$.

- (a) Compute the autocovariance function of ϵ_t^2 . Prove that ϵ_t^2 follows a weak ARMA model whose orders will be given.
- (b) Propose a moment estimator for μ_2 and b^2 . Show that this estimator is consistent.
- (c) When they exist, compute the matrices

$$I = \lim_{n \rightarrow \infty} \text{Var} \left\{ \sqrt{n} \begin{pmatrix} \frac{\partial Q_n(b_0, \mu_{02})}{\partial b_2} \\ \frac{\partial Q_n(b_0, \mu_{02})}{\partial \mu_2} \end{pmatrix} \right\}$$

and

$$J = \lim_{n \rightarrow \infty} \begin{pmatrix} \frac{\partial^2 Q_n(b_0, \mu_{02})}{\partial b_2^2} & \frac{\partial^2 Q_n(b_0, \mu_{02})}{\partial b_2 \partial \mu_2} \\ \frac{\partial^2 Q_n(b_0, \mu_{02})}{\partial \mu_2 \partial b_2} & \frac{\partial^2 Q_n(b_0, \mu_{02})}{\partial \mu_2^2} \end{pmatrix}.$$

What moment condition is necessary for the existence of I and J ?

- (d) Give the scheme of proof which would establish that, under some assumptions, the least-squares estimator $\hat{\theta}$ of the parameter $\theta_0 = (b_0, \mu_{02})'$ satisfies

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Sigma(\theta_0)),$$

where $\Sigma(\theta_0) = J^{-1} I J^{-1}$.

- (e) Let

$$\ell_t(\theta) = \log h_t + \frac{\epsilon_t^2}{h_t}, \quad h_t = \mu_2(1 + b\epsilon_{t-1})^2.$$

When it exists, compute

$$J^{QML} = E \left(\frac{\partial^2 \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right)$$

and show that

$$J^{QML} = E \left(\frac{1}{h_t^2(\theta_0)} \frac{\partial h_t(\theta_0)}{\partial \theta} \frac{\partial h_t(\theta_0)}{\partial \theta'} \right) = (\kappa_\eta - 1) \text{Var} \frac{\partial \ell_t(\theta_0)}{\partial \theta},$$

where

$$\kappa_\eta = \frac{\mu_{04}}{\mu_{02}^2}.$$

What moment condition is necessary for the existence of J^{QML} ?

- (f) Give the scheme of proof which would establish that, under some assumptions, the quasi-maximum likelihood estimator $\hat{\theta}^{QML}$ satisfies

$$\sqrt{n}(\hat{\theta}^{QML} - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Sigma(\theta_0)^{QML}).$$

- (g) What is the particular form of $\Sigma^{QML}(\theta_0)$ at $\theta_0 = (0, \mu_{02})'$? What are the consequences on the asymptotic properties of the estimator?
- (h) Compare $\Sigma(\theta_0)$ and $\Sigma^{QML}(\theta_0)$ at $\theta_0 = (0, \mu_{02})'$.
- (i) Without giving detailed proofs, extend the stationarity and QML estimation results to the model

$$\epsilon_t = \eta_t + b_1 \eta_t \epsilon_{t-1} + \cdots + b_q \eta_t \epsilon_{t-q}, \quad t \in \mathbb{Z}.$$

SOLUTION

1. (a) In view of the Cauchy root test, it suffices to show that almost surely

$$\lim_{i \rightarrow \infty} |b^i \eta_t \cdots \eta_{t-i}|^{1/i} < 1.$$

By the law of large numbers, this limit is equal to

$$\lim_{i \rightarrow \infty} \exp \left\{ \frac{1}{i} \left(\sum_{k=1}^i \log |b \eta_{t-k}| + \log |\eta_t| \right) \right\} = \exp \{ E \log |b \eta_t| \},$$

which shows the result.

- (b) For all n , we have

$$Z_{t,n} = \eta_t + b \eta_t Z_{t-1,n-1}.$$

Taking the limit, $Z_t = \eta_t + b \eta_t Z_{t-1}$, which shows that $(\epsilon_t) = (Z_t)$ is a nonanticipative solution of (D.6). Since $Z_t = f(\eta_t, \eta_{t-1}, \dots)$ (where $f : \mathbb{R}^\infty \rightarrow \mathbb{R}$ is measurable) and (η_t) is ergodic and stationary, (Z_t) is also stationary and ergodic.

- (c) We have $E \log |\eta_t / \sqrt{\mu_2}| = -0.635181$. The stationarity condition is thus written as

$$E \log \left| b \sqrt{\mu_2} \frac{\eta_t}{\sqrt{\mu_2}} \right| = \log |b \sqrt{\mu_2}| + E \log \left| \frac{\eta_t}{\sqrt{\mu_2}} \right| < 0,$$

or equivalently

$$|b| \sqrt{\mu_2} < \exp \{0.635181\} = 1.88736.$$

2. (a) For $n < m$, we have

$$E \left\{ \sum_{i=n}^m b^i \eta_t \cdots \eta_{t-i} \right\}^2 = \sum_{i=n}^m b^{2i} \mu_2^{i+1} \rightarrow 0$$

as $n, m \rightarrow \infty$ (that is, the sequence is a Cauchy sequence) if and only if

$$b^2 \mu_2 < 1.$$

- (b) If $b^2 \mu_2 < 1$ then, using the Jensen inequality, we have

$$E \log |b \eta_t| = \frac{1}{2} E \log b^2 \eta_t^2 \leq \frac{1}{2} \log E b^2 \eta_t^2 = \frac{1}{2} \log b^2 \mu_2 < 0.$$

- (c) When $b^2\mu_2 < 1$, we have seen that $(Z_{t,n})_n$ is a Cauchy sequence. Thus it converges in L^2 to some limit $\tilde{Z}_t$. It also converges almost surely to Z_t . Thus $Z_t = \tilde{Z}_t$ almost surely, and $EZ_t^2 < \infty$. To show the uniqueness of the solution, assume the existence of two second-order stationary solutions (Z_t) and (Z_t^*) . Then, for any $n \geq 1$,

$$Z_t - Z_t^* = b\eta_t(Z_{t-1} - Z_{t-1}^*) = b^n\eta_t\eta_{t-1}\cdots\eta_{t-n+1}(Z_{t-n} - Z_{t-n}^*).$$

By the Cauchy–Schwarz and triangular inequalities,

$$E|Z_t - Z_t^*| \leq |b|^n\mu_2^{n/2} \{ \|Z_1\|_2 + \|Z_1^*\|_2 \}.$$

Since this is true for all n , the condition $b^2\mu_2 < 1$ entails $E|Z_t - Z_t^*| = 0$, which implies $Z_t = Z_t^*$ almost surely.

- (d) If ϵ_t is a second-order stationary solution then

$$E\epsilon_t^2 = \mu_2 + b^2\mu_2 E\epsilon_t^2,$$

that is,

$$(1 - b^2\mu_2) E\epsilon_t^2 = \mu_2.$$

If $b^2\mu_2$ were greater than 1, the left-hand side of the previous inequality would be negative, which is impossible because the right-hand side is strictly positive.

3. (a) Such a solution is nonanticipative and satisfies

$$\begin{aligned} E\epsilon_t &= E\eta_t + bE\eta_t E\epsilon_{t-1} = 0, \\ E\epsilon_t^2 &= \frac{\mu_2}{1 - b^2\mu_2}, \\ \text{Cov}(\epsilon_t, \epsilon_{t-h}) &= 0, \quad \forall h > 0. \end{aligned}$$

It is thus a white noise. Let us show that (ϵ_t^2) is an ARMA process. Using the independence between η_t and ϵ_{t-k} and $E(\eta_t^2) = \mu_2$, we have, for $k > 0$,

$$\begin{aligned} \text{Cov}(\epsilon_t^2, \epsilon_{t-k}^2) &= \text{Cov}(\eta_t^2 + 2b\eta_t^2\epsilon_{t-1} + b^2\eta_t^2\epsilon_{t-1}^2, \epsilon_{t-k}^2) \\ &= 2b\text{Cov}(\eta_t^2\epsilon_{t-1}, \epsilon_{t-k}^2) + b^2\text{Cov}(\eta_t^2\epsilon_{t-1}^2, \epsilon_{t-k}^2) \\ &= 2b\mu_2\text{Cov}(\epsilon_{t-1}, \epsilon_{t-k}^2) + b^2\mu_2\text{Cov}(\epsilon_{t-1}^2, \epsilon_{t-k}^2). \end{aligned}$$

For $k > 1$,

$$\text{Cov}(\epsilon_{t-1}, \epsilon_{t-k}^2) = E(\epsilon_{t-1} \epsilon_{t-k}^2) = E(\eta_{t-1}(1 + b\epsilon_{t-2})\epsilon_{t-k}^2) = 0.$$

It follows that, for $k > 1$,

$$\text{Cov}(\epsilon_t^2, \epsilon_{t-k}^2) = b^2 \mu_2 \text{Cov}(\epsilon_{t-1}^2, \epsilon_{t-k}^2),$$

which shows that (ϵ_t^2) admits an ARMA(1, 1) representation. Finally, (ϵ_t) admits a weak GARCH(1, 1) representation.

(b) The volatility of the model is

$$\mu_2(1 + b\epsilon_{t-1})^2 = \mu_2 + b^2 \mu_2 \epsilon_{t-1}^2 + 2b\mu_2 \epsilon_{t-1},$$

whereas it is of the form

$$\omega + \alpha \epsilon_{t-1}^2$$

for an ARCH(1). The sign of ϵ_{t-1} is thus important. If $b < 0$, a negative return ϵ_{t-1} increases the volatility more than it does the return $-\epsilon_{t-1} > 0$. Such an asymmetry in the shocks is observed in real series, but is not taken into account by standard GARCH models.

4. (a) Since h_{0t} is the conditional expectation of ϵ_t^2 , ν_t is the strong innovation of ϵ_t^2 .
- (b) The process $\{\nu_t(h_{0t} - h_t)\}_t$ is ergodic and stationary, by arguments already used. The ergodic theorem entails that

$$\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{t=2}^n \nu_t(h_{0t} - h_t) = E(\eta_t^2 - \mu_{02}) E h_{0t}(h_{0t} - h_t) = 0 \quad \text{a.s.}$$

because $h_{0t}(h_{0t} - h_t)$ is independent of $(\eta_t^2 - \mu_{02})$, as measurable function of $\{\eta_u, u \leq t-1\}$.

- (c) We have $E(h_{0t} - h_t)^2 = 0$ if and only if

$$h_{0t} - h_t = (\mu_{02}b_0^2 - \mu_2b^2)\epsilon_{t-1}^2 + 2(\mu_{02}b_0 - \mu_2b)\epsilon_{t-1} + (\mu_{02} - \mu_2) = 0 \quad \text{a.s.}$$

This second-order equation in ϵ_{t-1} (or in ϵ_t by stationarity) admits a solution if and only if the coefficients are null, that is, if and only if $b = b_0$ and $\mu_2 = \mu_{02}$.

(d) Using the two last questions, almost surely,

$$\begin{aligned}\lim_{n \rightarrow \infty} Q_n(b, \mu_2) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=2}^n \{\epsilon_t^2 - h_{0t} + h_{0t} - h_t\}^2 \\ &= \lim_{n \rightarrow \infty} Q_n(b_0, \mu_{02}) + E(h_{0t} - h_t)^2 + 0 \\ &\geq \lim_{n \rightarrow \infty} Q_n(b_0, \mu_{02})\end{aligned}$$

with equality if and only if $b = b_0$ and $\mu_2 = \mu_{02}$. This suggests looking for a value of (b, μ_2) minimizing the criterion $Q_n(b, \mu_2)$. This is the least-squares method.

(e) If η_t is $\mathcal{N}(0, \mu_{02})$ distributed then the distribution of ϵ_t given $\{\epsilon_u, u < t\}$ is $\mathcal{N}(0, h_{0t})$. Given the initial value ϵ_1 , the quasi-log likelihood of $\epsilon_2, \dots, \epsilon_n$ is thus

$$\mathcal{L}_n(b, \mu_2) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{t=2}^n \left\{ \log h_t(b, \mu_2) + \frac{\epsilon_t^2}{h_t(b, \mu_2)} \right\}.$$

A quasi-maximum likelihood estimator satisfies

$$(\hat{b}, \hat{\mu}_2) = \arg \max_{(b, \mu_2) \in \Theta} \mathcal{L}_n(b, \mu_2),$$

where $\Theta \subset \mathbb{R} \times]0, \infty[$ is the parameter space. If Θ is a compact set, since the criterion is continuous, there always exists at least one QMLE.

Problem 7

Geweke (1986), Pantula (1986) and Milhøj (1987) introduced independently the log-GARCH(1, 1) model

$$\begin{cases} \epsilon_t &= \sigma_t \eta_t, \\ \ln \sigma_t^2 &= \omega + \alpha \ln \epsilon_{t-1}^2 + \beta \ln \sigma_{t-1}^2 \end{cases}$$

where ω , α and β are unknown coefficients, and (η_t) is a strong white noise such that $E\eta_0^2 = 1$ and $P(\eta_0 = 0) = 0$. This model, that we call "ordinary" log-GARCH, has not yet received much success among practitioners and academics because it is not directly suitable for series with possibly zero returns. The zero probability $P(\epsilon_t = 0)$ being not necessarily zero in practice, Sucarrat and Grønneberg (2016) recently proposed the "zero-adjusted" log-GARCH(1,1) model, whose a simple version is defined as follow.

Let z_t be a strong white noise such that $Ez_0^2 = 1$ and $P(z_0 = 0) = \pi \in [0, 1)$, and

$$\begin{cases} \epsilon_t &= \sigma_t z_t, \\ \ln \sigma_t^2 &= \omega + \alpha 1_{z_{t-1} \neq 0} \ln \epsilon_{t-1}^2 + \beta \ln \sigma_{t-1}^2 \end{cases}$$

where $1_{z_{t-1} \neq 0} \ln \epsilon_{t-1}^2$ is equal to zero when $z_{t-1} = 0$ (or equivalently when $\epsilon_{t-1} = 0$), and is equal to $\ln \epsilon_{t-1}^2$ otherwise.

1. INTERPRETATION OF THE LOG-GARCH

- (a) Recall the standard GARCH(1,1) formulation. Explain the difference between the standard and log-GARCH formulations in terms of positivity constraint on the coefficients.
- (b) Is the log-GARCH able to produce volatility clustering? for which values of the parameters?
- (c) Explain why the constraint $P(\eta_t = 0) = 0$ is necessary for the ordinary log-GARCH.
- (d) What happens with the ordinary log-GARCH when one value of the observed returns is zero, say $\epsilon_{t_0} = 0$?

2. EXISTENCE OF A STATIONARY SOLUTION

- (e) Show that for the ordinary log-GARCH, $\log \sigma_t^2$ satisfies an autoregressive equation. Give a necessary and sufficient condition for the existence of a strictly stationary and non anticipative solution to the ordinary log-GARCH(1,1) equation.

- (f) Show that the zero-adjusted log-GARCH(1,1) model satisfies

$$\log \sigma_t^2 = \omega(z_{t-1}) + a(z_{t-1}) \log \sigma_{t-1}^2,$$

for some functions $\omega(\cdot)$ and $a(\cdot)$ to be defined. Deduce that

$$E \log |\omega(z_1)| < \infty \quad \text{and} \quad (1 - \pi) \log |\alpha + \beta| + \pi \log |\beta| < 0$$

are sufficient conditions for the existence of a strictly stationary solution.

- (g) Give conditions on the parameters and on the distribution of z_t that entail $E|\log \sigma_t^2| < \infty$.

3. ESTIMATION Let $\epsilon_1, \dots, \epsilon_n$ be observations of the zero-adjusted log-GARCH(1,1) model. Denote by $\theta_0 = (\omega_0, \alpha_0, \beta_0)$ the true value of the unknown parameters $\theta = (\omega, \alpha, \beta)$. Define $\tilde{\sigma}_t^2(\theta)$ for $t = 1, \dots, n$ by

$$\log \tilde{\sigma}_1^2(\theta) = \omega, \quad \log \tilde{\sigma}_t^2(\theta) = \omega + \alpha 1_{\epsilon_{t-1} \neq 0} \ln \epsilon_{t-1}^2 + \beta \ln \tilde{\sigma}_{t-1}^2(\theta) \text{ for } t = 2, \dots, n.$$

Consider the estimator

$$\hat{\theta}_n = \arg \min_{\theta} Q_n(\theta), \quad Q_n(\theta) = \frac{1}{n} \sum_{t=1}^n \frac{\epsilon_t^2}{\tilde{\sigma}_t^2(\theta)} + \log \tilde{\sigma}_t^2(\theta).$$

- (h) What is the name of the estimator $\hat{\theta}_n$? what is the behavior of this estimator when it is applied to a standard GARCH model?
- (i) Give a condition that ensures the invertibility of the model, *i.e.* a condition that allows to approximate $\tilde{\sigma}_t^2(\theta)$ by a function $\sigma_t^2(\theta)$ which is measurable with respect to $\{\epsilon_u, u < t\}$, such that the process $\{\sigma_t(\theta)\}_t$ is stationary and ergodic.
- (j) Show that, under regularity conditions that will not be detailed, $Q_n(\theta)$ converges almost surely to some number $Q_\infty(\theta)$. What is the sign of $Q_\infty(\theta) - Q_\infty(\theta_0)$ when $\theta \neq \theta_0$?
- (k) In order to assess the finite-sample performance of the estimator we simulated $N = 1000$ independent trajectories of size $n = 500$ and $n = 2000$ of a log-GARCH(1,1) with $\pi = 0$ or $\pi = 0.1$. Tables D.1-D.2 summarize the results of the simulation experiments. Comment these tables.
- (l) Propose an estimator and a confidence bound for π .

Table D.1: Empirical distribution of 1000 estimations $\hat{\theta}_n$ when $\theta_0 = (0, 0.05, 0.90)$ and $\pi = 0$

	$n = 500$			$n = 2000$		
	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$
Minimum	-1.678950	0.001365	-0.4340	-0.086881	0.03176	0.8008
1st Quantile	-0.042604	0.043193	0.8526	-0.014883	0.04528	0.8828
Median	-0.012481	0.052776	0.8839	-0.002614	0.05006	0.8969
Mean	-0.036965	0.053865	0.8645	-0.005570	0.05051	0.8948
3rd Quantile	0.007941	0.063611	0.9062	0.006005	0.05514	0.9080
Maximum	0.057119	0.153684	0.9949	0.033447	0.07791	0.9377

Table D.2: Empirical distribution of 1000 estimations $\hat{\theta}_n$ when $\theta_0 = (0, 0.05, 0.90)$ and $\pi = 0.1$

	$n = 500$			$n = 2000$		
	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$
Minimum	-1.568258	-0.02926	-0.9272	-0.0535082	0.02237	0.7839
1st Quantile	-0.016044	0.04083	0.8496	0.0007615	0.04470	0.8830
Median	0.003348	0.05073	0.8900	0.0091188	0.04997	0.8965
Mean	-0.017393	0.05247	0.8596	0.0072426	0.05049	0.8952
3rd Quantile	0.017576	0.06308	0.9155	0.0153086	0.05609	0.9101
Maximum	0.052619	0.14458	1.0000	0.0435478	0.08520	0.9623

SOLUTION

1. (a) With the standard GARCH(1,1) model $\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ and, contrary to the log-GARCH, the parameters of the volatility are constrained to be positive.
- (b) If α is positive and β is close to 1, a large value of $\log \sigma_t^2$ will tend to be followed by other large values of the log-volatility, which entails volatility clustering.
- (c) This condition is necessary, otherwise ϵ_{t-1}^2 might be equal to zero and the volatility could not be computed.
- (d) The model is not compatible with zero observations. In that case the volatility can not be computed.
2. (e) We have

$$\log \sigma_t^2 = \omega + \alpha \log \eta_{t-1}^2 + (\alpha + \beta) \log \sigma_{t-1}^2 = \sum_{i=0}^{\infty} (\alpha + \beta)^i (\omega + \alpha \log \eta_{t-i-1}^2)$$

if and only if $|\alpha + \beta| < 1$.

- (f) We have

$$\begin{aligned} \log \sigma_t^2 &= \omega + \alpha 1_{z_{t-1} \neq 0} \log z_{t-1}^2 + (\alpha 1_{z_{t-1} \neq 0} + \beta) \log \sigma_{t-1}^2 \\ &:= \omega(z_{t-1}) + a(z_{t-1}) \log \sigma_{t-1}^2 \\ &= \omega(z_{t-1}) + a(z_{t-1}) \omega(z_{t-2}) + a(z_{t-1}) a(z_{t-2}) \omega(z_{t-3}) + \dots \end{aligned}$$

where, by the Cauchy rule, the series converges absolutely if

$$E \log |\omega(z_1)| < \infty \text{ and } E \log |a(z_{t-1})| = (1 - \pi) \log |\alpha + \beta| + \pi \log |\beta| < 0.$$

- (g) In view of the previous question, the conditions

$$|\alpha + \beta|(1 - \pi) + |\beta|\pi < 1 \quad \text{and} \quad E|1_{z_{t-1} \neq 0} \ln z_{t-1}^2| < \infty$$

entail $E|\log \sigma_t^2| < \infty$.

3. (h) This is the quasi-maximum likelihood estimator (QMLE), which coincides with the maximum likelihood estimator when the iid noise is Gaussian, but is generally consistent and asymptotically normal under less restrictive assumptions.

- (i) When $|\beta| < 1$ and (ϵ_t) is stationary and ergodic, one can define the stationary and ergodic process satisfying

$$\log \sigma_t^2(\theta) = \omega + \alpha 1_{\epsilon_{t-1} \neq 0} \ln \epsilon_{t-1}^2 + \beta \ln \sigma_{t-1}^2(\theta) \text{ for all } t.$$

We have $|\log \tilde{\sigma}_t^2(\theta) - \log \sigma_t^2(\theta)| \leq K|\beta|^t$ for all $t \geq 1$ and K a random variable depending on $\{\epsilon_u, u < 0\}$.

- (j) Showing that the limit of $Q_n(\theta)$ is the same as that of

$$\frac{1}{n} \sum_{t=1}^n \frac{\epsilon_t^2}{\sigma_t^2(\theta)} + \log \sigma_t^2(\theta),$$

the ergodic theorem shows that $Q_\infty(\theta) = E \frac{\epsilon_1^2}{\sigma_1^2(\theta)} + \log \sigma_1^2(\theta)$, provided the expectation exists. Using the inequality $\log x \leq x - 1$, we have

$$Q_\infty(\theta) - Q_\infty(\theta_0) = E \left\{ \frac{\sigma_1^2(\theta_0)}{\sigma_1^2(\theta)} + \log \frac{\sigma_1^2(\theta)}{\sigma_1^2(\theta_0)} - 1 \right\} \geq 0$$

with equality if and only if $\sigma_1^2(\theta) = \sigma_1^2(\theta_0)$ a.s.

- (k) The QMLE behaves as expected: the distribution of the estimates is more concentrated around the true values when $n = 2000$ than when $n = 500$. It seems that the estimator (in particular that of β) is a bit more accurate when $\pi = 0$ (Table D.1) than when $\pi = 0.1$ (Table D.2).
- (l) Since $\eta_t = 0$ if and only if $\epsilon_t = 0$, one can propose the estimator

$$\hat{\pi} = \frac{1}{n} \sum_{t=1}^n 1_{\{\epsilon_t=0\}}.$$

Using the normal approximation

$$\frac{\hat{\pi} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \approx \mathcal{N}(0, 1),$$

we obtain the confidence interval

$$\left[\hat{\pi} - \Phi^{-1} \left(1 - \frac{\alpha}{2} \right) \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}, \hat{\pi} + \Phi^{-1} \left(1 - \frac{\alpha}{2} \right) \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} \right]$$

at the asymptotic confidence level $1 - \alpha$.

Problem 8

Exercise 1: Recall that the autocorrelation function $\rho(\cdot)$ of a second-order stationary ARMA(p, q) process satisfies

$$\rho(h) = \sum_{i=1}^p \phi_i \rho(h-i), \quad h > q,$$

where the ϕ_i are the AR coefficients. Denote by B the lag operator: $BX_t = X_{t-1}$. Let (ϵ_t) be a strictly stationary solution of a GARCH(p, q) model such that $E(\epsilon_t^4) < \infty$.

1. Show that (ϵ_t^2) admits an ARMA representation. What is the relation satisfied by the autocorrelation function ρ_{ϵ^2} of this process?
2. The aim of this question is to check that the function ρ_{ϵ^2} has positive values.
 - (a) Show the property when $p = q = 1$.
 - (b) Using the ARMA representation of ϵ_t^2 , show that there exist constants c_i and a noise ν_t such that

$$\epsilon_t^2 = c_0 + \sum_{i=1}^{\infty} c_i \nu_{t-i}.$$

- (c) Let P and Q be two polynomials with positive coefficients such that $P(0) = Q(0) = 0$. Assume that $1 - Q(z) = 0$ implies that $|z| > 1$. Show by induction that $\{1 - Q(B)\}^{-1}P(B)$ is a series in B with positive terms.
 - (d) Prove that the coefficients c_i are positive.
 - (e) Deduce the property.
3. The aim of this question is to show that the function ρ_{ϵ^2} is not always decreasing.
 - (a) Verify that for a GARCH(1, 1) model the function ρ_{ϵ^2} is decreasing.
 - (b) Assume that $q = 2$ and $p = 0$, that is, $\epsilon_t = \sqrt{\omega + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2} \eta_t$.
 - i. What is the relationship between $\rho_{\epsilon^2}(h)$, $\rho_{\epsilon^2}(h-1)$ and $\rho_{\epsilon^2}(h-2)$ for $h > 0$?
 - ii. Deduce an expression for $\{\rho_{\epsilon^2}(1)\}^{-1} \rho_{\epsilon^2}(2)$ as a function of α_1 and α_2 .

- iii. Prove that for some set of values of α_1 and α_2 , the function ρ_{ϵ^2} is not decreasing.

Exercise 2: Consider the ARCH(q) model

$$\begin{cases} \epsilon_t &= \sigma_t \eta_t, \\ \sigma_t^2 &= \omega_0 + \sum_{i=1}^q \alpha_{0i} \epsilon_{t-i}^2, \end{cases}$$

where $\omega_0 > 0$, $\alpha_{0i} \geq 0$, $i = 1, \dots, q$, and (η_t) is an iid sequence such that $E(\eta_t) = 0$, and $\text{Var}(\eta_t) = 1$. Let $\epsilon_1, \dots, \epsilon_n$ be n observations of the process (ϵ_t) and let $\epsilon_0, \dots, \epsilon_{1-q}$ be initial values. Introduce the vector $Z_{t-1} \in \mathbb{R}^q$ defined by $Z'_{t-1} = (1, \epsilon_{t-1}^2, \dots, \epsilon_{t-q}^2)$, and the $n \times q$ matrix X and the $n \times 1$ vector Y given by

$$X = \begin{pmatrix} 1 & \epsilon_0^2 & \dots & \epsilon_{-q+1}^2 \\ \vdots & & & \\ 1 & \epsilon_{n-1}^2 & \dots & \epsilon_{n-q}^2 \end{pmatrix} = \begin{pmatrix} Z'_0 \\ \vdots \\ Z'_{n-1} \end{pmatrix}, \quad Y = \begin{pmatrix} \epsilon_1^2 \\ \vdots \\ \epsilon_n^2 \end{pmatrix}.$$

Denote by $\theta_0 = (\omega_0, \alpha_{01}, \dots, \alpha_{0q})'$ the true value of the parameter.

1. Show that the OLS estimator $\hat{\theta} = (\hat{\omega}, \hat{\alpha}_1, \dots, \hat{\alpha}_q)'$ of θ_0 is given by

$$\hat{\theta} = (X'X)^{-1}X'Y.$$

We use the notation $\sigma_t^2(\hat{\theta}) = \hat{\omega} + \sum_{i=1}^q \hat{\alpha}_i \epsilon_{t-i}^2$ and $\tilde{\epsilon}_t = \{\sigma_t(\hat{\theta})\}^{-1} \epsilon_t$.

2. Give conditions ensuring the following almost sure convergences as $n \rightarrow \infty$:

$$\frac{1}{n} X'X \rightarrow E_{\theta_0}(Z_{t-1}Z'_{t-1}), \quad \frac{1}{n} X'Y \rightarrow E_{\theta_0}(Z_{t-1}\epsilon_t^2).$$

3. Prove that $\hat{\theta}$ converges almost surely to θ_0 .
4. In order to take into account the conditional heteroscedasticity, define the weighted least-squares estimator

$$\tilde{\theta} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y},$$

where

$$\tilde{X} = \begin{pmatrix} \sigma_1^{-2}(\hat{\theta}) & \tilde{\epsilon}_0^2 & \dots & \tilde{\epsilon}_{-q+1}^2 \\ \vdots & & & \\ \sigma_n^{-2}(\hat{\theta}) & \tilde{\epsilon}_{n-1}^2 & \dots & \tilde{\epsilon}_{n-q}^2 \end{pmatrix}, \quad \tilde{Y} = \begin{pmatrix} \tilde{\epsilon}_1^2 \\ \vdots \\ \tilde{\epsilon}_n^2 \end{pmatrix}.$$

- (a) Without going into all the mathematical details, justify the introduction of such an estimator.
- (b) Show that

$$\sqrt{n}(\tilde{\theta} - \theta_0) = \left(\frac{1}{n} \sum_{t=1}^n \frac{Z_{t-1} Z'_{t-1}}{\{\sigma_t(\hat{\theta})\}^4} \right)^{-1} \left\{ \frac{1}{\sqrt{n}} \sum_{t=1}^n \frac{Z_{t-1}(\eta_t^2 - 1)}{\{\sigma_t(\hat{\theta})\}^2} \right\}. \quad (\text{D.7})$$

- (c) Let $J = E(\sigma_t^{-4} Z_{t-1} Z'_{t-1})$. Justify the following results:

$$\frac{1}{n} \sum_{t=1}^n \sigma_t^{-4} Z_{t-1} Z'_{t-1} \rightarrow J, \quad \text{a.s.},$$

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \sigma_t^{-2} Z_{t-1} (\eta_t^2 - 1) \xrightarrow{\mathcal{L}} \mathcal{N}(0, (\mu_4 - 1)J).$$

- (d) Assume that the asymptotic distribution of the right-hand side of (D.7) does not change when $\sigma_t^2(\hat{\theta})$ is replaced by σ_t^2 . Deduce the asymptotic distribution of $\sqrt{n}(\tilde{\theta} - \theta_0)$.

Problem 9

For any random variable X , denote

$$X^+ = \max\{X, 0\}, \quad X^- = \min\{X, 0\}.$$

Thus $X^+ \geq 0$ and $X^- \leq 0$ almost surely. Consider the threshold GARCH(1, 1) model, or TGARCH(1, 1), defined by

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t = \omega + \alpha_+ \epsilon_{t-1}^+ - \alpha_- \epsilon_{t-1}^- + \beta \sigma_{t-1}, \end{cases} \quad (\text{D.8})$$

where (η_t) is a centred iid sequence with variance 1, and where ω is strictly positive, and α_+ , α_- and β are nonnegative numbers. In model (D.8), the parameter β is called the shock persistence parameter. In order to introduce some asymmetry for this persistence parameter, that is, a different value when $\epsilon_{t-1} < c\sigma_{t-1}$ and when $\epsilon_{t-1} > c\sigma_{t-1}$ for some constant c (that is, depending on whether the price fell abnormally or not), consider the model

$$\begin{cases} \epsilon_t = \sigma_t \eta_t \\ \sigma_t = \omega + \alpha_+ (\epsilon_{t-1} - c\sigma_{t-1})^+ - \alpha_- (\epsilon_{t-1} - c\sigma_{t-1})^- + \beta \sigma_{t-1}. \end{cases} \quad (\text{D.9})$$

1. Explain briefly the difference between the TGARCH model defined by (D.8) and the standard GARCH(1, 1) model, and why the TGARCH model might be of interest for financial series modeling.
2. Rewrite (D.9) to introduce an asymmetric persistence parameter. Why might such a model be of interest?
3. Study of the TGARCH model defined by (D.8)
 - (a) Give expressions for ϵ_t^+ and ϵ_t^- as functions of σ_t , η_t^+ and η_t^- .
 - (b) Show that $\sigma_t = \omega + a(\eta_{t-1})\sigma_{t-1}$, where a is a function that will be specified.
 - (c) Give a sufficient condition for the existence of a nonanticipative and ergodic strictly stationary solution.
 - (d) Specify this stationarity condition when $\beta = 0$ and η_t is $\mathcal{N}(0, 1)$ distributed.
 - (e) Give a necessary condition for the existence of a nonanticipative stationary solution such that $E\sigma_t < \infty$. Give $E\sigma_t$, when it exists.
 - (f) Give a necessary condition for the existence of a nonanticipative stationary solution such that $E\sigma_t^2 < \infty$. Give $E\sigma_t^2$, when it exists.

- (g) Assume that η_t has a symmetric distribution. When they exist, give the almost sure limits of

$$\frac{1}{n} \sum_{t=1}^n \epsilon_t^+ \epsilon_{t-1}^+ \quad \text{and} \quad \frac{1}{n} \sum_{t=1}^n \epsilon_t^+ \epsilon_{t-1}^-$$

as $n \rightarrow \infty$. Give a simple empirical method for checking if $\alpha_+ < \alpha_-$ (do not go into the details of the test).

4. Study of the model defined by (D.9)

- (a) Give a sufficient condition for the existence of a nonanticipative and ergodic strictly stationary solution.
- (b) Give a *necessary and sufficient* condition for the existence of a nonanticipative stationary solution such that $E\sigma_t < \infty$.
- (c) Assume that η_t has a strictly positive density on $\mathbb{R}$. Show that, except in the degenerate case where $\alpha_+ = \alpha_- = 0$, the model is identifiable, that is, denoting the ‘true’ value of the parameter by $\sigma_t = \sigma_t(\theta)$, where $\theta = (\omega, \alpha_+, \alpha_-, c, \beta)$, we have

$$\sigma_t(\theta) = \sigma_t(\theta^*) \text{ a.s. if and only if } \theta = \theta^*.$$

- (d) Give a method for estimating the parameter θ .

SOLUTION

1. In the standard GARCH formulation, the conditional variance $\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ does not depend on the sign of ϵ_{t-1} . In the TGARCH formulation with $\alpha_- > \alpha_+$, a negative return ϵ_{t-1} entails a greater increase in the volatility than a positive return of the same magnitude. Empirical studies have shown the existence of such asymmetries in most financial series.
2. We have

$$\sigma_t = \begin{cases} \omega + \alpha_+\epsilon_{t-1} + (\beta - c\alpha_+)\sigma_{t-1} & \text{if } \eta_{t-1} \geq c, \\ \omega - \alpha_-\epsilon_{t-1} + (\beta + c\alpha_-)\sigma_{t-1} & \text{if } \eta_{t-1} \leq c. \end{cases}$$

In this model the persistence coefficient is equal to $(\beta - c\alpha_+)\sigma_{t-1}$ when $\eta_{t-1} \geq c$ (that is, when $\epsilon_{t-1} \geq c\sigma_{t-1}$), and $(\beta + c\alpha_-)\sigma_{t-1}$ when $\eta_{t-1} \leq c$. A motivation for considering this model is that a negative shock should increase volatility more than a positive one of the same amplitude, and also that this increase should have a longer effect. By testing the assumption that $c = 0$, one can test whether the persistence of the negative shocks is the same as that of the positive shocks.

3. (a) The positivity of the coefficients guarantees that $\sigma_t \geq 0$ (starting from an initial value $\sigma_0 \geq 0$). We thus have $\epsilon_t^+ = \sigma_t\eta_t^+$ and $\epsilon_t^- = \sigma_t\eta_t^-$.
- (b) In view of (a), $\sigma_t = \omega + a(\eta_{t-1})\sigma_{t-1}$, where $a(\eta) = \alpha_+\eta^+ - \alpha_-\eta^- + \beta$.
- (c) For all $n \geq 1$, let

$$s_t(n) = \omega \left\{ 1 + \sum_{i=1}^n \prod_{k=1}^i a(\eta_{t-k}) \right\}.$$

We have $s_t(n) = \omega + a(\eta_{t-1})s_{t-1}(n-1)$ for all n . If $s_t = \lim_{n \rightarrow \infty} s_t(n)$ exists, then the previous equality still holds true at the limit, and the solution is given by $\sigma_t = s_t$ (stationarity and ergodicity follow from the fact that $s_t = f(\eta_{t-1}, \eta_{t-2}, \dots)$, where f is a measurable function and (η_t) is ergodic and stationary). Since all the terms involved in $s_t(n)$ are positive, s_t is an increasing limit which exists in $\mathbb{R}^+ \cup \{+\infty\}$. By the Cauchy root criterion, the limit exists in $\mathbb{R}^+$ if

$$\lambda := \limsup_{n \rightarrow \infty} \left\{ \prod_{k=1}^n a(\eta_{t-k}) \right\}^{1/n} < 1.$$

By the law of large numbers, $\log \lambda = E \log a(\eta_1)$. The condition $E \log a(\eta_1) < 0$ thus guarantees the existence of the solution.

- (d) When $\beta = 0$ and the distribution of η_1 is symmetric, we have

$$\begin{aligned} E \log a(\eta_1) &= \int_0^\infty \log(\alpha_+ x) dP_\eta(x) - \int_{-\infty}^0 \log(\alpha_- x) dP_\eta(x) \\ &= \frac{1}{2} \log(\alpha_+ \alpha_-) + E \log |\eta_1|. \end{aligned}$$

The condition is then given by $\alpha_+ \alpha_- < \exp(-2E \log |\eta_1|)$.

- (e) If there exists a nonanticipative stationary solution such that $E\sigma_t < \infty$, then

$$E\sigma_t = \omega + Ea(\eta_t)E\sigma_t = \frac{\omega}{1 - \alpha_+ E\eta_t^+ + \alpha_- E\eta_t^- - \beta},$$

but this is possible only if $Ea(\eta_t) < 1$, that is, $\alpha_+ E\eta_1^+ - \alpha_- E\eta_1^- + \beta < 1$.

- (f) If there exists a nonanticipative stationary solution such that $E\sigma_t^2 < \infty$, then

$$\begin{aligned} E\sigma_t^2 &= \omega^2 + Ea^2(\eta_t)E\sigma_t^2 + 2\omega Ea(\eta_t)E\sigma_t \\ &= \left(1 + 2\frac{Ea(\eta_t)}{1 - Ea(\eta_t)}\right) \frac{\omega^2}{1 - Ea^2(\eta_t)}, \end{aligned}$$

which is possible only if $Ea^2(\eta_1) < 1$.

- (g) By the ergodic theorem, and noting that $\epsilon_t^+ \epsilon_t^- = 0$, under the assumption that the moments exist we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n \epsilon_t^+ \epsilon_{t-1}^+ = E\eta_t^+ E(\omega + \alpha_+ \epsilon_{t-1}^+ + \beta \sigma_{t-1}) \epsilon_{t-1}^+$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n \epsilon_t^+ \epsilon_{t-1}^- = E\eta_t^+ E(\omega - \alpha_- \epsilon_{t-1}^- + \beta \sigma_{t-1}) \epsilon_{t-1}^-, \quad \text{a.s.}$$

Since the distribution of η_t is symmetric, we have

$$E\epsilon_t^+ = E\eta_t^+ E\sigma_t = -E\epsilon_t^-, \quad E\epsilon_t^+ \sigma_t = -E\epsilon_t^- \sigma_t$$

and $E(\epsilon_t^+)^2 = E(\epsilon_t^-)^2 = E\sigma_t^2/2$. For testing $\alpha_+ < \alpha_-$ one can thus use the statistic $n^{-1} \sum_{t=1}^n \epsilon_t^+ \epsilon_{t-1}$, which should converge to $E(\eta_t^+)(\alpha_+ - \alpha_-)E\sigma_t^2/2$.

4. (a) The arguments of part 3(c) show that a condition for the existence of such a solution is $E \log b(\eta_1) < 0$ with

$$\begin{aligned} b(\eta_t) &= (\alpha_+ \eta_t + \beta - c\alpha_+) \mathbf{1}_{\eta_t \geq c} + (-\alpha_- \eta_t + \beta + c\alpha_-) \mathbf{1}_{\eta_t < c} \\ &= \alpha_+ (\eta_t - c)^+ - \alpha_- (\eta_t - c)^- + \beta. \end{aligned}$$

- (b) The arguments of part 3(e) show that the condition $Eb(\eta_1) < 1$ is necessary. By Jensen's inequality, $E \log b(\eta_1) \leq \log Eb(\eta_1)$. The condition $Eb(\eta_1) < 1$ is thus sufficient for the existence of a strictly stationary solution of the form

$$\sigma_t = \omega \{1 + b(\eta_t) + b(\eta_t)b(\eta_{t-1}) + \cdots\}.$$

By Beppo Levi's theorem, this solution satisfies

$$E\sigma_t = \frac{\omega}{1 - Eb(\eta_1)} < \infty.$$

- (c) Let $\theta^* = (\omega^*, \alpha_+^*, \alpha_-^*, c^*, \beta^*)$ and consider the case $c^* \leq c$. The case $c^* \geq c$ is handled similarly. Denote by R_t any measurable function with respect to $\sigma\{\epsilon_u, u \leq t\}$. We have

$$\sigma_t = \alpha_+ \sigma_{t-1} (\eta_{t-1} - c) \mathbf{1}_{\{\eta_{t-1} > c\}} - \alpha_- \sigma_{t-1} (\eta_{t-1} - c) \mathbf{1}_{\{\eta_{t-1} < c\}} + R_{t-2},$$

and, writing $\sigma_t^* = \sigma_t(\theta^*)$,

$$\begin{aligned} \sigma_t^* &= \alpha_+^* (\sigma_{t-1} \eta_{t-1} - c^* \sigma_{t-1}^*) \mathbf{1}_{\{\sigma_{t-1} \eta_{t-1} > c^* \sigma_{t-1}^*\}} \\ &\quad - \alpha_-^* (\sigma_{t-1} \eta_{t-1} - c^* \sigma_{t-1}^*) \mathbf{1}_{\{\sigma_{t-1} \eta_{t-1} < c^* \sigma_{t-1}^*\}} + R_{t-2}. \end{aligned}$$

Assume that $\sigma_t = \sigma_t^*$ almost surely (for any t). Then we have

$$\begin{aligned} \eta_{t-1} (\alpha_+ - \alpha_+^*) \mathbf{1}_{\{\eta_{t-1} > c\}} &= R_{t-2}, \\ \eta_{t-1} (-\alpha_- - \alpha_+^*) \mathbf{1}_{\{c^* < \eta_{t-1} < c\}} &= R_{t-2}, \\ \eta_{t-1} (-\alpha_- + \alpha_-^*) \mathbf{1}_{\{\eta_{t-1} < c^*\}} &= R_{t-2}. \end{aligned}$$

This implies that $\alpha_+ = \alpha_+^*$, $\alpha_- = \alpha_-^*$, and $\alpha_+^* = -\alpha_-$ if $c^* < c$. Since $\alpha_+^* \neq -\alpha_-$, we have $c = c^*$, $\alpha_+ = \alpha_+^*$ and $\alpha_- = \alpha_-^*$. We then have $\omega - \omega^* + (\beta - \beta^*) \eta_{t-1} R_{t-2} = 0$ a.s., which entails $\omega = \omega^*$ and $\beta = \beta^*$.

- (d) One could estimate θ by the quasi-maximum likelihood method, that is, with the aid of the estimator

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \sum_{t=1}^n \left\{ \frac{\epsilon_t^2}{\tilde{\sigma}_t^2(\theta)} + \log \tilde{\sigma}_t^2(\theta) \right\},$$

where Θ is a compact parameter space which constrains the parameters to be positive, and $\tilde{\sigma}_t^2(\theta)$ is defined recursively by

$$\tilde{\sigma}_t(\theta) = \begin{cases} \omega + \alpha_+ \epsilon_{t-1} + (\beta - c\alpha_+) \tilde{\sigma}_{t-1}(\theta) & \text{if } \epsilon_{t-1} \geq c\tilde{\sigma}_{t-1}(\theta) \\ \omega - \alpha_- \epsilon_{t-1} + (\beta + c\alpha_-) \tilde{\sigma}_{t-1}(\theta) & \text{if } \epsilon_{t-1} \leq c\tilde{\sigma}_{t-1}(\theta), \end{cases}$$

with for instance $\tilde{\sigma}_0(\theta) = \omega$ as initial value.

Problem 10

Let the model

$$X_t = c(\Delta_t) + \eta_t, \quad (\text{D.10})$$

where (η_t) is an i.i.d. sequence of variables, with $E(\eta_t) = 0$ and $\text{Var}(\eta_t) = 1$, $c(\cdot)$ is a real valued function, and (Δ_t) is a Markov chain on $\{1, \dots, d\}$ which is supposed stationary, irreducible, aperiodic and independent of (η_t) . Let $p(i, j) = P(\Delta_t = j \mid \Delta_{t-1} = i)$, let $\mathbb{P} = (p(i, j))$ the transition matrix and let $\pi(i)$ the stationary probabilities of the chain.

1. Briefly justify the interest of this model.
2. Is the process (X_t) strictly and second-order stationary? a detailed proof is expected.
3. Derive the autocorrelation function ρ of (X_t) and show that $\rho(k) \rightarrow 0$ as $k \rightarrow \infty$.
4. Let ξ_t the $d \times 1$ vector whose i th component is equal to 1 if $\Delta_t = i$, and to zero otherwise. Show that (ξ_t) is a Markov chain. After defining a state space for this chain, write the transition probabilities.
5. Show that, for some matrix A we have

$$E(\xi_{t+1} \mid \xi_t) = A\xi_t.$$

Deduce that

$$\xi_{t+1} = A\xi_t + u_{t+1},$$

where $E(u_{t+1} \mid \xi_t) = 0$. Show that the sum of the components of u_{t+1} is equal to 0. Compute the matrix $\text{Var}(u_{t+1} \mid \xi_t)$.

6. Show that Model (D.10) admits a state-space representation, with state vector (ξ_t) . Does this representation satisfy the properties required to apply the Kalman filter?
7. We have observations $X_1, \dots, X_n$ generated from Model (D.10) where $\eta_t \sim \mathcal{N}(0, 1)$.
 - (a) Write the vector θ of the unknown parameters.

- (b) Let $g_k(X_1, \dots, X_k | \Delta_k = i)$ the density of the first k observations given $\Delta_k = i$, for $k \geq 1$, and let

$$F_k(i) = g_k(X_1, \dots, X_k | \Delta_k = i) \pi(i).$$

Establish a formula relating $F_k(i)$ and the $F_{k-1}(j)$, $j = 1, \dots, d$. Deduce a recursive relation allowing to compute the vectors F_k , where $F_k = (F_k(1), \dots, F_k(d))'$.

- (c) Deduce an expression of the likelihood allowing to estimate θ .
- (d) A statistician who does not know the true model estimates an AR(1) model: $X_t = \phi X_{t-1} + u_t$ by least-squares. Let $\hat{\phi}_n$ denote the least-squares estimator of the parameter ϕ . Does $\hat{\phi}_n$ converge when $n \rightarrow \infty$? Using this estimator, how could the error of model specification be detected?

SOLUTION

1. This type of modelling is useful for series displaying changes in levels. Contrary to break models, the model allows the series to return to the different regimes. Another difference is that the model is compatible with stationarity.
2. X_t is a function of (Δ_t, η_t) , which is a strictly and second-order stationary process because (Δ_t) and (η_t) are both stationary and are independent. Therefore (X_t) is strictly and second-order stationary.
3. We have, for $k > 0$, using the independence between (Δ_t) and (η_t) ,

$$\begin{aligned}
 \text{Cov}(X_t, X_{t-k}) &= \text{Cov}(c(\Delta_t), c(\Delta_{t-k})) \\
 &= \sum_{i,j=1}^d c(i)c(j)p^{(k)}(i,j)\pi(i) - \left(\sum_{i=1}^d c(i)\pi(i) \right)^2 \\
 &= \sum_{i,j=1}^d c(i)c(j)\pi(i)\{p^{(k)}(i,j)\pi(i) - \pi(j)\} \rightarrow 0 \quad \text{as } k \rightarrow \infty,
 \end{aligned}$$

because the k -step transition converges to the stationary probability. We also have

$$V(X_t) = V\{c(\Delta_t)\} + V(\eta_t) = \sum_{i=1}^d c^2(i)\pi(i) - \left(\sum_{i=1}^d c(i)\pi(i) \right)^2 + 1.$$

The autocorrelation is $\rho(k) = \text{Cov}(X_t, X_{t-k})/V(X_t)$.

4. We have $\xi_t = f(\Delta_t)$ so (ξ_t) is a Markov Chain. The state space is $\{e_1, \dots, e_d\}$ where (e_i) denote the canonical basis of $\mathbb{R}^d$. Moreover, $P(\xi_t = e_j | \xi_{t-1} = e_i) = p(i, j)$.
5. We have

$$E(\xi_{t+1} | \xi_t) = \begin{pmatrix} P(\Delta_{t+1} = 1 | \xi_t) \\ \vdots \\ P(\Delta_{t+1} = d | \xi_t) \end{pmatrix} = \begin{pmatrix} p(1, 1) \mathbf{1}_{\Delta_t=1} + \dots + p(d, 1) \mathbf{1}_{\Delta_t=d} \\ \vdots \\ p(1, d) \mathbf{1}_{\Delta_t=1} + \dots + p(d, d) \mathbf{1}_{\Delta_t=d} \end{pmatrix} = \mathbb{P}' \xi_t.$$

By definition of the conditional expectation, u_{t+1} satisfies $E(u_{t+1} | \xi_t) = 0$. Let $e = (1, \dots, 1)'$. We have

$$e' u_{t+1} = e' \xi_{t+1} - e' \mathbb{P}' \xi_t = 1 - e' \xi_t = 0.$$

6. Equation (1) can be written $X_t = c'\xi_t + \eta_t$ with $c' = (c(1), \dots, c(d))$. Together with the equation $\xi_t = \mathbb{P}'\xi_{t-1} + u_t$, this constitutes a state-space model. Independence and normality of the noises are crucial to apply the Kalman filter: both are non satisfied here.
7. (a) $\theta = (p(1, 1), \dots, p(1, d-1), p(2, 1), \dots, p(d-1, d), c(1), \dots, c(d))'$.
 (b) Exactly as was done in the course we get, for $k > 0$,

$$F_k(i) = \phi(X_k - c(i)) \sum_{j=1}^d F_{k-1}(j)p(j, i), \quad i = 1, \dots, d,$$

where ϕ is the density of the $\mathcal{N}(0, 1)$. The sequence $(F_k(i), i = 1, \dots, d)_{k=1, \dots, n}$ can be computed recursively (with initial values for the $F_0(i)$).

- (c) The likelihood is $L_n(\theta) = \sum_{i=1}^d F_n(i)$.
- (d) We have $\hat{\phi}_n = \frac{\sum_{t=1}^n X_t X_{t-1}}{\sum_{t=1}^n X_{t-1}^2}$, which converges to the first-order autocorrelation of (X_t) . If we consider the residuals $X_t - \hat{\phi}_n X_{t-1}$, there empirical mean will be generally far from 0 (even for large n).

Additional References (to be inserted into the general list)

- Chou, R.Y. (2005) Forecasting Financial Volatilities with Extreme Values: The Conditional Autoregressive Range (CARR) Model. *Journal of Money, Credit and Banking* 37, 561–582.
- Sucarrat, G., and Grønneberg, S. (2016). Models of Financial Return With Time-Varying Zero Probability. <https://mpra.ub.uni-muenchen.de/id/eprint/68931>